



BANARAS HINDU UNIVERSITY

B.Sc.(Hons.) Courses

Effective from 2024 under NEP

Offered by

Faculty of Science



BANARAS HINDU UNIVERSITY



Department of Mathematics

Faculty of Science

SYLLABI

B.Sc. (Hons.) Programme

In

Mathematics

DEPARTMENT OF MATHEMATICS
BANARAS HINDU UNIVERSITY, VARANASI-221005
Syllabus
NEP Course Structure for Four Year Undergraduate Mathematics Program

Ist Year

Semester –I			
Course Code	Course Description	Title	Credits
MATMJ11	MAT-Major (4)	Calculus	4
MATMN11	MAT-Minor*(4)	Calculus	4
MATSE11	MAT-Skill Enhancement#(SE) (3)	Ethics in Academia and Mathematical Exploration	3
MATMD11	Multidisciplinary**(MD) (3)	Basic Mathematics-I	3
VA1+VA2+AE1 (2+2+2)		From the Pool of VA and AE Courses	6
Total credit			20

Semester –II			
Course Code	Course Description	Title	Credits
MATMJ21	MAT-Major (4)	Algebra	4
MATMN21	MAT-Minor* (4)	Algebra	4
MATSE21	MAT-Skill Enhancement#(SE) (3)	LaTeX Typesetting for Beginners	3
MATMD21	Multidisciplinary**(MD) (3)	Basic Mathematics-II	3
VA3+VA4+AE2 (2+2+2)		From the Pool of VA and AE Courses	6
Total credit			20

IInd Year

Semester –III			
Course Code	Course Description	Title	Credits
MATMJ31	MAT-Major (8)	Linear Algebra	4
MATMJ32		Analysis	4
MATMV31	MAT-Minor(VOC)* (4)	Python for Mathematical Applications	4
MATSE31	MAT-Skill Enhancement#(SE)(3)	Sage Math for Beginners	3
MATMD31	Multidisciplinary**(MD) (3)	Basic Mathematics-III	3
Ability Enhancement (AE3) (2)		From the Pool AE Courses	2
Total credit			20

Semester –IV			
Course Code	Course Description	Title	Credits
MATMJ41	MAT-Major (14)	Calculus of several variables	4
MATMJ42		Vector and Tensor Analysis	3
MATMJ43		Differential Equations	4
MATMJ44		Mechanics	3
MATMN41	MAT-Minor* (4)	Differential Equations	4
Ability Enhancement (AE3) (2)		From the Pool AE Courses	2
Total credit			20

IIIrd Year

Semester –V			
Course Code	Course Description	Title	Credits
MATMJ51	MAT-Major (14)	Abstract Algebra	4
MATMJ52		Metric Spaces	3
MATMJ53		Analytic Geometry	3
MATMJ54		Numerical Analysis	3+1(p)
MATMV51	MAT-Minor(Voc)* (4)	Optimization Techniques	4
MAT-IN-51	Internship(2)	Internship to be done during vacation	2
Total credit			20

Semester –VI			
Course Code	Course Description	Title	Credits
MATMJ61	MAT-Major (16)	Measure and integration	3
MATMJ62		Complex Analysis	3
MATMJ63		Differential Geometry	4
MATMJ64		Number Theory	3
		Optional (opt any one)	
MATMJ65		Linear Programming and Applications	3
MATMJ66		Special Theory of Relativity	3
MATMJ67		Space Dynamics	3
MATMJ68		Discrete Mathematics	3
MATMJ69		Mathematical Modelling	3
MATMJ610		Dynamical System	3
MATMJ611		Data Structures	2+1P
MATMJ612		Analytical Visualization	1+2P
MATMJ613		Graph Theory	3
MATMJ614		Vedic Mathematics	3
MATMV61	MAT-Minor (Voc)* (4)	Formal Languages and Automata Theory	4
Total credit			20

IVth Year

Semester –VII			
Course Code	Course Description	Title	Credits
MATMJ71	MAT-Major (16)	Topology	4
MATMJ72		Advanced Algebra	4
MATMJ73		Functional Analysis	4
MATMJ74		Differentiable Manifolds	4
MATMN71	MAT-Minor A*(4)	Topology	4
Total credit			20

Semester –VIII (Category-1: For Mathematics Honours only)			
Course Code	Course Description	Title	Credits
MATMJ81	MAT-Major (16) (opt any four papers)	Geometric Topology	4X4=16
MATMJ82		Fourier Analysis	
MATMJ83		Nonlinear Programming	
MATMJ84		Quadratic Programming	
MATMJ85		Fluid Dynamics	
MATMJ86		Numerical Analysis	
MATMJ87		Mathematical Modeling	
MATMJ88		Numerical Solution of Partial Differential Equations	
MATMJ89		Riemannian Manifolds	
MATMJ810		Information and Coding Theory	
MATMJ811		Algebraic Geometry	
MAT-Minor A*(4)		Any one of the above	
Total credit			20

Semester –VIII (Category-2: For Mathematics Honours only)			
Course Code	Course Description	Title	Credits
MATMJ8R1	MAT-Major (4) (opt any onepaper)	Geometric Topology	4
MATMJ8R2		Fourier Analysis	
MATMJ8R3		Nonlinear Programming	
MATMJ8R4		Quadratic Programming	
MATMJ8R5		Fluid Dynamics	
MATMJ8R6		Numerical Analysis	
MATMJ8R7		Mathematical Modeling	
MATMJ8R8		Numerical Solution of Partial Differential Equations	
MATMJ8R9		Riemannian Manifolds	
MATMJ8R10		Information and Coding Theory	
MATMJ8R11		Algebraic Geometry	
MAT-Minor A*(4)	Any one of the above		4
Dissertation (12)	MAT-DS-8R1	Research Dissertation	12
Total credit			20

Course Title	MATMJ11/MATMN11: Calculus			
Credits & Hours of Teaching		Theory	Practical	Cumulative
	Credits	4	0	4
	Hours of Teaching (Total)	60	0	60
Course Objectives (not more than 100 words)	This course aims to provide a rigorous understanding of sequences and various tests for the convergence of series. It delves into the foundational concepts of limit and continuity, including the algebra of limits, continuous functions, and uniform continuity. Students will study differentiability and essential theorems such as the Intermediate Value Theorem, Mean Value Theorem, and Taylor's Theorem. Additionally, the course covers curvature, asymptotes, and the tracing of plane curves, integrating these concepts with practical applications in integral calculus and the geometry of curves.			
Units	Course Content			Teaching Hrs
I	Series: Sequences and their limits. Theorems on limits. Convergent and divergent sequences. Monotone Sequences. Subsequences and the Bolzano-Weierstrass Theorem.			12
II	Series: The Cauchy Criterion. Convergent, Divergent, and Oscillatory series, Comparison test, D'Alembert's ratio test. Raabe's test. Cauchy's nth root test.			12
III	Limit and Continuity: Limits of functions, Algebra of limits. Continuity and uniform continuity of functions, Intermediate value theorem. Differentiability. Chain rule of differentiation.			12
IV	Darboux's theorem, Rolle's theorem, and Mean value theorems. Leibnitz theorem. Taylor's and Maclaurin's theorems. Maxima and Minima of functions. Nowhere differentiable functions.			12
V	Curvature. Asymptotes. Concavity and Convexity: Point of Inflection, Singular Points. Tracing of plane curves (Cartesian and polar). Definite Integral as the limit of sum.			12
Texts / References	1. T.M. Apostol: Calculus, Vol I, John Wiley and Sons, Inc., New York, 1967. 2. R.G. Bartle; D.R. Sherbert: Introduction to Real Analysis, John Wiley and Sons, Inc., New York, 2011. 3. S.R. Ghorpade; B.V. Limaye: A Course in Calculus and Real Analysis, 2nd Edition, Springer, 2018. 4. Amber Habib, Calculus, Cambridge University Press, 2024. 5. G. Prasad: Differential Calculus, Pothishala Pvt. Ltd. Allahabad, 2000. 6. G. Prasad: Integral Calculus, Pothishala Pvt. Ltd. Allahabad, 2000. 7. A.N. Singh: The Theory and Construction of Non-differentiable Functions, Lucknow University Studies No. I, Newul Kishore Press, Lucknow, 1935. 8. N. Saran: Calculus, Pothishala Pvt. Ltd. Allahabad, 1963. 9. W.F. Trench: Introduction to Real Analysis, Pearson Education, 2003.			
Learning Outcomes (not more than 100 words)	Upon completing this course, students can rigorously analyze sequences and series, determining their limits and convergence using various theorems and tests. They will understand and apply the concepts of limits, continuity, and differentiability to functions, utilizing key theorems such as Intermediate and Mean Value Theorems. Students will also learn to evaluate curvature, identify asymptotes, and trace plane curves. Additionally, they will develop the ability to apply integral calculus in understanding the geometric properties of curves and solve related mathematical problems.			

Course Title		MATSE11: Ethics in Academia and Mathematical Exploration			
Credits &Hours of Teaching		Theory	Practical	Cumulative	
	Credits	3	0	3	
	Hours of Teaching (Total)	45	0	45	
Course Objectives (not more than 100 words)	The objective of this course is to equip students with a robust understanding of ethical practices in higher education and academic research. Simultaneously, it aims to deepen their appreciation for key mathematical concepts and theorems. Students will engage in discussions about the implications of ethical decision-making in research, while also exploring various mathematical topics, such as irrational numbers, logarithms, and the Fibonacci sequence. By the end of the course, students will be able to apply ethical reasoning to academic challenges and recognize the beauty and significance of mathematics in real-world contexts.				
Units	Course Content				Teaching Hrs
I	Ethics in Higher Education and Academic Research. Ethics of Publication. Ethical Practices in Science Outreach. Ethical Issues Associated with Gender-Bias.				15
II	Six proofs of the infinity of primes. Binomial coefficients are (almost) never powers. Representing numbers as sums of two squares. Three applications of Euler's formula. Sets, functions, and the continuum hypothesis. In praise of inequalities.				15
III	Some irrational numbers. The golden mean and other roots. e : The queen of growth and decay. Logarithms. Infinite series: Fractals and the myth of forever. Fibonacci Numbers.				15
Texts / References	1. A. Ghosh, A.K. Singhvi, K. Muralidhar (Editors) Ethics in Science Education, Research and Governance, INSA, 2019. 2. M. Aigner and M. Ziegler: Proofs from THEBOOK, Springer, 2001. 3. A. Khare and A. Lachowska, Beautiful, Simple, Exact, Crazy: Mathematics in the Real World, Yale University Press, 2015. 4. J. Matousek, Thirty-three Miniatures: Mathematical and Algorithmic Applications of Linear Algebra, American Mathematical Society, 1991.				
Learning Outcomes (not more than 100 words)	Upon completing this course, students will have a comprehensive understanding of the ethical frameworks governing higher education and academic research. They will critically analyze issues such as publication ethics, gender bias, and science outreach practices. Additionally, students will explore fundamental mathematical concepts, including the infinity of primes, binomial coefficients, and Euler's formula. They will develop the ability to connect ethical considerations with mathematical principles, fostering a holistic view of the interconnectedness between ethics and the pursuit of knowledge in both academia and mathematics.				

Course Title	MATMD11: Basic Mathematics-I				
Credits & Hours of Teaching		Theory	Practical	Cumulative	
	Credits	3	0	3	
	Hours of Teaching (Total)	45	0	45	
Course Objectives (not more than 100 words)	This course aims to comprehensively understand matrix theory, linear algebra, and difference equations. Students will learn to perform matrix operations, solve linear systems, and analyze sequences and their limits. The course also covers applying mathematical concepts to real-world problems, including pharmacokinetics and interpreting rates of change in various contexts.				
Units	Course Content				Teaching Hrs
I	Matrices, matrix addition, and multiplication. Determinants. Elementary row and column operations, Echelon form, rank of a matrix. The inverse of a matrix. Solution of a system of linear equations using matrices and determinants.				15
II	Geometric and Arithmetic Sequences, Monotone Sequences, Limit of a Sequence, Discrete Difference Equations, Linear Difference Equation with Constant Coefficients, Introduction to Pharmacokinetics.				15
III	Limit of a Function, Limit Properties, Right and Left Limits, Continuity, Intermediate Value Theorem, Average Rate of Change, Estimating Rates of Change for Data, Velocity, Examples of Rates of Change.				15
Texts / References	1. Erin N. Bodine, Suzanne Lenhart, and Louis J. Gross, Mathematics for the Life Sciences, Princeton University Press, 2014. 2. Robert G. Bartle and Donald R. Sherbert, Introduction to Real Analysis, John Wiley and Sons, Inc., New York, 2011.				
Learning Outcomes (not more than 100 words)	By the end of the course, students will be able to solve systems of linear equations using matrix methods and understand and apply matrix operations, determinants, and inverses. They will also be proficient in determining the rank of matrices and solving linear difference equations. Additionally, students will be able to analyze sequences and series, apply limits and continuity concepts, and interpret rates of change in various contexts, including pharmacokinetics.				

Course Title	MATMD11: Basic Mathematics-I				
Credits & Hours of Teaching		Theory	Practical	Cumulative	
	Credits	3	0	3	
	Hours of Teaching (Total)	45	0	45	
Course Objectives (not more than 100 words)	This course aims to comprehensively understand matrix theory, linear algebra, and difference equations. Students will learn to perform matrix operations, solve linear systems, and analyze sequences and their limits. The course also covers applying mathematical concepts to real-world problems, including pharmacokinetics and interpreting rates of change in various contexts.				
Units	Course Content				Teaching Hrs
I	Matrices, matrix addition, and multiplication. Determinants. Elementary row and column operations, Echelon form, rank of a matrix. The inverse of a matrix. Solution of a system of linear equations using matrices and determinants.				15
II	Geometric and Arithmetic Sequences, Monotone Sequences, Limit of a Sequence, Discrete Difference Equations, Linear Difference Equation with Constant Coefficients, Introduction to Pharmacokinetics.				15
III	Limit of a Function, Limit Properties, Right and Left Limits, Continuity, Intermediate Value Theorem, Average Rate of Change, Estimating Rates of Change for Data, Velocity, Examples of Rates of Change.				15
Texts / References	1. Erin N. Bodine, Suzanne Lenhart, and Louis J. Gross, Mathematics for the Life Sciences, Princeton University Press, 2014. 2. Robert G. Bartle and Donald R. Sherbert, Introduction to Real Analysis, John Wiley and Sons, Inc., New York, 2011.				
Learning Outcomes (not more than 100 words)	By the end of the course, students will be able to solve systems of linear equations using matrix methods and understand and apply matrix operations, determinants, and inverses. They will also be proficient in determining the rank of matrices and solving linear difference equations. Additionally, students will be able to analyze sequences and series, apply limits and continuity concepts, and interpret rates of change in various contexts, including pharmacokinetics.				

Course Title		MATMJ21: Algebra			
Credits & Hours of Teaching		Theory	Practical	Cumulative	
	Credits	4	0	4	
	Hours of Teaching (Total)	60	0	60	
Course Objectives (not more than 100 words)	This course aims to comprehensively understand real and complex numbers, including their roots and geometrical representations. Students will explore polynomials, their roots, the Fundamental Theorem of Algebra, and the relationships between roots and coefficients. The course covers Newton's method for integral roots, equivalence relations, and the division algorithm in $\mathbb{Z}$, including divisibility and the Euclidean algorithm. Students will learn about the Fundamental Theorem of Arithmetic, modular arithmetic, and basic congruence properties. The course also introduces group theory, covering basic properties, symmetries, dihedral groups, orders, subgroups, cyclic groups, and their classifications.				
Units	Course Content				Teaching Hrs
I	Real and complex numbers, roots of complex numbers, the n th roots of unity, and geometrical representation.				12
II	Polynomials, roots of polynomial, Fundamental theorem of algebra, Relations between the roots and the coefficients, imaginary roots, integral and rational roots; Newton's method for integral roots.				12
III	Relation, equivalence relation, and equivalence class. Division algorithm in $\mathbb{Z}$, Divisibility, and the Euclidean algorithm, Fundamental theorem of arithmetic, Modular arithmetic, and basic properties of congruence.				12
IV	Groups, Basic properties of groups, Symmetries of a square, Dihedral group, Order of a group, Order of an element, Subgroups, Centre of a group, Centralizer of an element.				12
V	Cyclic groups and properties, Generators of a cyclic group, Classification of subgroups of cyclic groups.				12
Texts / References	1. W.S. Burnside, and A.W. Panton, The Theory of Equations, Vol. 1. Eleventh Edition, Dover Publications, Inc., 1979. 2. David M. Burton. Elementary Number Theory, 7th Edition, McGraw-Hill Education, 2011. 3. Leonard Eugene Dickson, First Course in the Theory of Equations. John Wiley & Sons, Inc., 2009. 4. I.N. Herstein, Topics in Algebra John Wiley and Sons, 2nd edition, 2004. 5. D.S. Malik, J.M. Mordeson and M.K. Sen, Fundamentals of Abstract Algebra, The McGraw-Hill Companies, Inc., 1997.				
Learning Outcomes (not more than 100 words)	Upon completing this course, students can analyze and solve problems involving real and complex numbers, roots of polynomials, and the n th roots of unity. They will understand and apply the Fundamental Theorem of Algebra and Newton's method. Students will gain proficiency in equivalence relations, divisibility, the Euclidean algorithm, and modular arithmetic. Additionally, they will develop a solid foundation in group theory, including understanding the properties of groups, dihedral groups, subgroups, cyclic groups, and the classification of subgroups of cyclic groups.				

Course Title		MATSE21: LaTeX Typesetting for Beginners			
Credits & Hours of Teaching		Theory	Practical	Cumulative	
	Credits	2	1	3	
	Hours of Teaching (Total)	30	20	50	
Course Objectives (not more than 100 words)	This course will equip students with the foundational skills to create professional-quality documents using LaTeX. By the end of the course, students will be proficient in formatting text, structuring documents, incorporating mathematical content, managing bibliographies, and customizing layouts. The course also introduces advanced topics like creating custom commands and using LaTeX for presentations, preparing students to handle various document types.				
Units	Course Content				Teaching Hrs
I	Introduction to LaTeX: Overview of LaTeX and its uses, Installing LaTeX (MikTeX/TeX Live) and an editor (TeXworks/Overleaf), Creating your first LaTeX document, Document structure: preamble, document body, and sections, Compiling LaTeX documents (PDF output) Text Formatting: Basic text formatting (bold, italics, underline, Fonts and sizes, Paragraph formatting (alignment, spacing, indentation, Lists: itemized, enumerated, and description lists, Special characters and symbols. Document Structure and Sections: Creating titles, sections, subsections, and chapters, Table of contents, lists of figures, and tables; page styles: headers, footers, and page numbers; Cross-referencing within the document.				10
II	Working with Mathematical Content: Basic mathematical symbols and equations, Display vs. inline math, Aligning equations, Common math environments (e.g., equation, align), Numbering and referencing equations. Tables and Figures: Creating tables: basic and complex tables, Adding captions and labels to tables, Inserting images and figures, Image formats and sizing, Positioning images within the text, Wrapping text around images				10
III	Bibliography and Citations: Introduction to referencing and bibliographies, Using BibTeX and Biber for bibliography management, Creating a bibliography: thebibliography environment vs. bib file, Common citation styles (APA, IEEE, etc.), Managing citations and references. Advanced Topics in LaTeX: Customizing document classes (e.g., article, report, book), Adding footnotes and endnotes, Creating custom commands and environments, Introduction to LaTeX packages, Presentation with LaTeX: Introduction to Beamer. Final Project and Review: Review all topics covered, Work on a final project (e.g., create a formatted document with text, equations, figures, and references), and Troubleshoot common LaTeX issues.				10
Texts / References	1. Donald E. Knuth, <i>The TEXbook</i>, Addison-Wesley Publishing Company, 1991. 2. Leslie Lamport, <i>A Document Preparation System</i>, Addison-Wesley Publishing Company, 1994. 3. Michael Spivak, <i>The Joy of TeX: A Gourmet Guide to Typesetting with the AMSTeX</i>, American Mathematical Society, 1990.				
Learning Outcomes (not more than 100 words)	Students can Create and compile LaTeX documents with structured content after completing this course. Format text, lists, and paragraphs, and manage document sections. Incorporate and format mathematical equations, tables, and figures. Manage references and bibliographies using BibTeX and Biber. Customize document layouts and styles and use LaTeX for presentations. Troubleshoot common issues and complete a final project demonstrating their LaTeX skills.				

Course Title	MATMD21: Basic Mathematics-II				
Credits & Hours of Teaching		Theory	Practical	Cumulative	
	Credits	3	0	3	
	Hours of Teaching (Total)	45	0	45	
Course Objectives (not more than 100 words)	This course aims to develop students' understanding of derivatives and their applications. Students will learn to compute derivatives of various functions, including exponential, trigonometric, and logarithmic. The course will cover techniques for finding maxima and minima, optimization problems, and the role of derivatives in analyzing function behavior. Key objectives include mastering differentiation rules, applying the Mean Value Theorem, and solving real-world problems using derivative concepts.				
Units	Course Content				Teaching Hrs
I	Concept of a Derivative, Limit Definition of a Derivative of a Function, Derivatives of Exponential Functions, Derivatives of Trigonometric Functions, Derivatives and Continuity, Derivatives of Logarithmic Functions.				15
II	The Chain Rule for the Composition of Functions, Quotient and Reciprocal Rules, Exponential Models, Higher Derivatives.				15
III	Maxima and Minima, First Derivative Test, Mean Value Theorem, Concavity, Optimization Problems.				15
Texts / References	1. Erin N. Bodine, Suzanne Lenhart, and Louis J. Gross, Mathematics for the Life Sciences, Princeton University Press, 2014. 2. William F. Trench, Introduction to Real Analysis, Pearson Education, 2003. 3. George B. Thomas, Ross L. Finney, Calculus and Analytic Geometry, 1998.				
Learning Outcomes (not more than 100 words)	By the end of the course, students will be proficient in calculating derivatives using various rules and techniques, including the limit definition, chain rule, and rules for exponential, trigonometric, and logarithmic functions. They can analyze functions for maxima, minima, and concavity, apply the Mean Value Theorem, and solve optimization problems. Additionally, students will understand how derivatives relate to function continuity and will be able to interpret and apply higher derivatives and exponential models.				

Course Title		MATMJ31: Linear Algebra			
Credits & Hours of Teaching		Theory	Practical	Cumulative	
	Credits	4	0	4	
	Hours of Teaching (Total)	60	0	60	
Course Objectives (not more than 100 words)	This course aims to provide a foundational understanding of linear algebra, focusing on vector spaces, subspaces, and linear transformations. Students will explore concepts such as bases, dimension, rank, nullity, and the properties of matrices. The course covers systems of linear equations, inner product spaces, orthogonality, orthogonal and unitary transformations, and the spectral properties of matrices. Additionally, students will learn about invariant subspaces, eigenvalues, eigenvectors, the Cayley-Hamilton Theorem, and various canonical forms. The course will also introduce bilinear forms, matrix representations, and quadratic forms with applications to the classification of curves and surfaces.				
Units	Course Content				Teaching Hrs
I	Vector spaces, Subspaces, Span, Bases, Dimension, sum and quotient of Vector spaces, Maps of vector spaces, Linear transformations, Isomorphism, Rank and nullity, Dual space, Transpose, and annihilator.				12
II	Matrix of a linear Transformation, similar matrices, change of basis, Rank of Matrix, Normal form, System of linear equations and their solutions.				12
III	Inner product spaces and their properties, orthogonality, and orthonormalization, Orthogonal complement Adjoint of a linear transformation.				12
IV	Orthogonal and Unitary transformations. Hermitian and skew Hermitian matrices. Invariant subspaces, Eigenvalues and eigenvectors, Characteristic polynomial, Cayley-Hamilton Theorem.				12
V	Diagonalization and triangularization, Cyclic Subspaces, Nilpotent transformations, Canonical Forms. Bilinear form, matrix representation, Quadratic forms, and application to classification of curves and surfaces.				12
Texts / References	1. Artin, M. : Algebra, Prentice-Hall, 1994. 2. Herstein, I. N. : Topics in Algebra, Wiley Eastern, 1987. 3. Hoffman, K. and Kunze, R. : Linear Algebra, Prentice-Hall, 1972. 4. Halmos, P. R. : Finite-Dimensional Vector Spaces, Springer-Verlag, 1993. 5. Kwak, Jin Ho; Hong, Sungpyo Linear algebra. Second edition. Birkhäuser Boston, Inc., Boston, MA, 2004.				
Learning Outcomes (not more than 100 words)	Upon completing this course, students can analyze and manipulate vector spaces and subspaces, determine linear dependence and independence, and understand bases and dimensions. They will be proficient in linear transformations, isomorphisms, and matrix representations, including similar matrices and change of basis. Students will solve systems of linear equations, work with inner product spaces, and apply concepts of orthogonality and orthonormalization. They will understand eigenvalues, eigenvectors, and the Cayley-Hamilton Theorem and perform diagonalization and triangularization. Additionally, students will apply bilinear and quadratic forms to classify curves and surfaces.				

Course Title		MATMJ32: Analysis			
Credits & Hours of Teaching		Theory	Practical	Cumulative	
	Credits	4	0	4	
	Hours of Teaching (Total)	60	0	60	
Course Objectives (not more than 100 words)	This course aims to provide an in-depth understanding of the real numbers, their algebraic and order properties, and the completeness property of $\mathbb{R}$. Students will explore sequences, focusing on limit superior and limit inferior, and study convergence theorems. The course covers series, including alternating and absolutely convergent series, with tests by Dirichlet and Abel. Additionally, the course thoroughly examines the Riemann Integral, integrability criteria, and the fundamental and mean value theorems of integral calculus. Students will also study improper integrals, convergence tests, and the application of integrals as functions of a parameter.				
Units	Course Content				Teaching Hrs
I	The Real numbers: The Algebraic and Order Properties of $\mathbb{R}$. Absolute Value and Real Line.				12
II	The Completeness Property of $\mathbb{R}$. Applications of the Supremum Property. Limit superior and limit inferior of sequences. Monotone convergence theorem.				12
III	Alternating series, absolute convergent series. Dirichlet's and Abel's test. Riemann Integral, Integrability of continuous and monotonic Functions.				12
IV	Fundamental theorem of integral calculus, Mean Value theorems of integral calculus.				12
V	Improper integrals and their convergence. Comparison test, Beta and Gamma functions, Abel's and Dirichlet's test, Integral as a function of a parameter and its applications.				12
Texts / References	1. T. M. Apostol, Mathematical Analysis, Narosa Publishing House, New Delhi, 1985. 2. Richard Dedekind, What Are Numbers and What Should They Be? (A Classic Work) (Revised, edited and translated by H. Pogorzelski, W. Ryan and W. Snyder, Research Institute for Mathematics, Orono, Maine, 1995. 3. R. R. Goldberg, Real Analysis, Oxford & IBH Publishing Company, New Delhi, 1970. 4. S. Lang, Undergraduate Analysis, Springer-Verlag, New York, 1983. 5. W. Rudin, Principles of Mathematical Analysis, McGraw-Hill, 2023.				
Learning Outcomes (not more than 100 words)	Upon completing this course, students will be able to understand and apply the algebraic and order properties of $\mathbb{R}$, as well as the completeness and supremum properties. They will analyze sequences using limit superior and inferior, and apply the monotone convergence theorem. Students will evaluate alternating and absolutely convergent series using Dirichlet's and Abel's tests. They will comprehend the Riemann Integral and its integrability criteria, apply integral calculus's fundamental and mean value theorems, and determine the convergence of improper integrals using various tests. Additionally, students will apply integrals as functions of a parameter to solve practical problems.				

Course Title	MATMV31: Python for Mathematical Applications				
Credits &Hours of Teaching		Theory	Practical	Cumulative	
	Credits	3	1	4	
	Hours of Teaching (Total)	45	20	65	
Course Objectives (not more than 100 words)	This course aims to provide students with a comprehensive introduction to Python programming. It covers fundamental concepts such as literals, variables, identifiers, data types, and control structures for managing program flow. Students will gain practical experience with Python's core data structures, including lists, tuples, dictionaries, and sets. The course also delves into using functions, focusing on creating reusable code through value-returning and non-value-returning functions, understanding parameter passing, return values, and variable scope.				
Units	Course Content				Teaching Hrs
I	Fundamental concepts: Literals, variables and identifiers, strings, operators, expressions and data types, Control structures				15
II	Boolean expressions, selection control, iterative control Lists: List structures, Lists, (sequences), iterating over lists; Changing, adding, removing elements, Tuples,				15
III	Dictionaries and sets Functions: Program routines, calling value-returning functions, calling non value-returning functions, parameter passing, return values, variable scope.				15
Texts / References	1. Eric Matthes, Python Crash Course A Hands on, Project based Introduction to Programming, 2016, William Pollock, No Starch Press, Inc. 2. Mark Lutz, Learning Python,O’reilly,2009. 3. S.Sridhar, J. Indumathi, V.M. Hariharan, Python Programming, Pearson, 2023.				
Learning Outcomes (not more than 100 words)	By the end of this course, students will Understand and apply fundamental Python programming concepts, including data types, variables, and operators. Use control structures such as loops and conditional statements to manage program flow effectively. Manipulate and operate on Python's core data structures, including lists, tuples, dictionaries, and sets. Develop and utilize functions to create modular, reusable code with an understanding of parameter passing and scope. Write Python programs that demonstrate problem-solving skills and effective use of Python’s features.				

Course Title		MATSE31: Sage Math for Beginners			
Credits &Hours of Teaching		Theory	Practical	Cumulative	
	Credits	2	1	3	
	Hours of Teaching (Total)	30	20	50	
Course Objectives (not more than 100 words)	This course introduces students to SageMath, a versatile open-source mathematics software system. The course covers the essential features of SageMath, enabling students to perform arithmetic, algebra, calculus, and linear algebra operations, as well as create and analyze graphs. By the end of the course, students will be capable of using SageMath for various mathematical applications, from basic computations to solving complex problems in fields like number theory and discrete mathematics.				
Units	Course Content				Teaching Hrs
I	Introduction to SageMath: Overview of SageMath: What is it and why use it?, Installing SageMath (locally or using SageMathCell/CoCalc)., Introduction to the SageMath interface and basic commands., Creating and running SageMath notebooks. Arithmetic operations and basic functions. Algebra with SageMath: Working with symbolic expressions and variables. Simplifying expressions, expanding, and factoring. Solving algebraic equations (linear, quadratic, polynomial). Working with systems of equations. Introduction to matrices and basic matrix operations. Functions and Graphing: Defining and evaluating functions in SageMath. Plotting functions: basic plots, multiple plots, and customizing plots. Exploring more complex graphs: polar plots, parametric plots, and 3D plots. Applications: modeling real-world phenomena with functions and graphs.				10
II	Calculus with SageMath: Limits and continuity. Differentiation: finding derivatives and applying differentiation rules. Integration: definite and indefinite integrals. Series and summation. Applications: solving calculus problems using SageMath. Number Theory and Discrete Mathematics: Introduction to number theory: primes, gcd, lcm, and modular arithmetic. Working with integers, rationals, and modular forms. Basic combinatorics: permutations, combinations, and binomial coefficients. Applications in cryptography and coding theory.				10
III	Linear Algebra in SageMath: Introduction to vectors and vector spaces. Matrix operations: addition, multiplication, and inversion. Solving linear systems using matrix methods. Eigenvalues and eigenvectors. Applications: solving linear algebra problems using SageMath. Advanced Topics in SageMath. Introduction to SageMath packages and libraries (e.g., NumPy, SciPy). Polynomial algebra and advanced algebraic structures. Basic differential equations: solving ODEs and PDEs. Introduction to statistics and probability using SageMath. Applications: using SageMath in research and complex problem-solving.Final Project and Review: Review of all topics covered. Working on a final project (e.g., solving a real-world problem using SageMath). Troubleshooting common issues in SageMath. Course feedback and further learning resources.				10
Texts / References	1. Gregory V. Bard, <i>SageMath for Undergraduates</i> (online version), 2010. 2. Paul Zimmermann, <i>Computational Mathematics with SageMath</i> , SIAM,2019.				
Learning Outcomes (not more than 100 words)	Upon completing this course, students can: Navigate the SageMath interface and perform basic arithmetic and algebraic operations. Utilize SageMath for graphing functions, performing calculus operations, and solving linear algebra problems. Apply SageMath in areas such as number theory, combinatorics, and cryptography. Use SageMath packages for advanced mathematical computations, including differential equations and statistics. Complete a final project demonstrating the practical application of SageMath to a real-world problem.				

Course Title	MATMD31: Basic Mathematics-III			
Credits & Hours of Teaching		Theory	Practical	Cumulative
	Credits	3	0	3
	Hours of Teaching (Total)	45	0	45
Course Objectives (not more than 100 words)	This course aims to thoroughly understand integration techniques and their applications, including finding areas, volumes, and average values. Students will learn to use substitution and integration by parts to evaluate integrals. The course will also introduce differential equations, focusing on methods such as the separation of variables, and apply these concepts to population dynamics and stability models. Key objectives include mastering integration techniques, applying them to real-world problems, and understanding differential equations and their practical implications.			
Units	Course Content			Teaching Hrs
I	Definition of an Integral, Antiderivatives, Fundamental Theorem of Calculus, Antiderivatives and Integrals, Average Values. Substitution Method, Integration by Parts. Definite Integral and its applications.			15
II	The Area between Two Curves, The Volume of a Solid of Revolution, Density Functions			15
III	Introduction to Differential Equations, Separation of Variables Method, Models of Limited Population Growth, Equilibria and Stability, Homeostasis			15
Texts / References	1. Erin N. Bodine, Suzanne Lenhart, and Louis J. Gross, Mathematics for the Life Sciences, Princeton University Press, 2014. 2. William F. Trench, Introduction to Real Analysis, Pearson Education, 2003. 3. George B. Thomas, Ross L. Finney, Calculus and Analytic Geometry, 1998.			
Learning Outcomes (not more than 100 words)	By the end of the course, students will be able to compute and interpret integrals and antiderivatives and apply the Fundamental Theorem of Calculus. They will use methods such as substitution and integration by parts to solve integrals and analyze applications, including the area between curves, volumes of solids of revolution, and density functions. Students will also understand and solve differential equations using the separation of variables method and apply these concepts to population growth, equilibria, and homeostasis models.			

Course Title		MATMJ41: Calculus of Several Variables			
Credits & Hours of Teaching		Theory	Practical	Cumulative	
	Credits	4	0	4	
	Hours of Teaching (Total)	60	0	60	
Course Objectives (not more than 100 words)	This course aims to provide a comprehensive understanding of calculus in Euclidean n -space. Students will explore Euclidean norms, distances, and tangent spaces and delve into the concepts of open sets and functions from $\mathbb{R}^n$ to $\mathbb{R}^m$. The course covers limits, continuity, partial and directional derivatives, gradients, and the Jacobian matrix. Students will learn about the Mean Value Theorem, Euler's Theorem, and the Inverse and Implicit Function Theorems. The course also includes second differentials, Taylor series, extrema, Lagrange multipliers, and integration techniques.				
Units	Course Content				Teaching Hrs
I	Euclidean n -space $\mathbb{R}^n$. Euclidean norm and Euclidean distance. Tangent spaces in $\mathbb{R}^n$. Open sets in $\mathbb{R}^n$.				12
II	Functions from $\mathbb{R}^n$ to $\mathbb{R}^m$. Limit of sequences in $\mathbb{R}^m$. Limit and continuity of functions from $\mathbb{R}^n$ to $\mathbb{R}^m$. Partial derivatives and directional derivatives. Gradient of a function. Derivatives of functions between Euclidean space spaces.				12
III	First differential and Jacobian matrix. Change of Variables. Chain Rule. Mean Value Theorem. Euler's Theorem. Inverse Function Theorem and Implicit Function Theorem (Constant Rank Theorem optional).				12
IV	Hessian matrix, Second differentials and Taylor Series. Maxima, Minima and Saddles. Lagrange Multipliers.				12
V	Integration: Fubini's Theorem. Applications to area, surface area and volume. Beta and Gamma functions. Dirichlet's theorem, Liouville's theorem.				12
Texts / References	1. T.M. Apostol: Mathematical Analysis, Narosa Publishing House, New Delhi, 1985. 2. W. Rudin: Principle of Mathematical Analysis (Thrid edition) McGraw-Hill Kogakusha, 1976, International Student Edition. 3. J.R. Munkres: Analysis on manifolds. Addison-Wesley Publishing Company, Advanced Book Program, Redwood City, CA, 1991. 4. C.H. Edwards: Advanced Calculus of Several Variables, Academic Press, 1973. 5. T. M. Apostol: Calculus, Volume 2, 2nd. Ed., Wiley India, 2007. 6. J.J. Callahan: Advanced Calculus - A Geometric View, Springer, 2010. 7. R. Courant; F. John: Introduction to Calculus and Analysis, Volume 2, John Wiley and Sons, 1974. 8. J. Cooper: A MATLAB Companion for Multivariable Calculus, Harcourt/Academic Press 2001. 9. J.R. Magnus; H. Neudecker: Matrix differential calculus with applications in statistics and econometrics. John Wiley and Sons, Third Ed. 2007. 10. M. Spivak: Calculus on manifolds. A modern approach to classical theorems of advanced calculus. W. A. Benjamin, Inc., New York-Amsterdam, 1965.				
Learning Outcomes (not more than 100 words)	Upon completing this course, students can analyze Euclidean n -space, compute Euclidean norms, and understand tangent spaces. They will evaluate the limits and continuity of multivariable functions and calculate partial and directional derivatives. Students will apply the chain rule, the Mean Value Theorem, Euler's Theorem, and the Inverse and Implicit Function Theorems. They will determine maxima, minima, and saddle points using second differentials and Taylor series and solve optimization problems with Lagrange multipliers. Additionally, students will apply Fubini's Theorem, evaluate Beta and Gamma functions, and use Green's, Stokes', and Gauss' theorems for line and surface integrals.				

Course Title		MATMJ42: Vector and Tensor Analysis			
Credits & Hours of Teaching		Theory	Practical	Cumulative	
	Credits	3	0	3	
	Hours of Teaching (Total)	45	0	45	
Course Objectives (not more than 100 words)	This course aims to provide a comprehensive understanding of differential operators and vector integration, focusing on gradient, divergence, and curl, as well as the Helmholtz decomposition. Students will explore line, surface, and volume integrals and apply integral theorems such as Green's theorem, Gauss's divergence theorem, and Stokes' theorem. The course covers curvilinear coordinates, including orthogonal systems, and extends these concepts to tensor analysis. Students will learn about contravariant and covariant tensors, coordinate transformations, physical laws, and the metric tensor, Christoffel symbols, geodesics, and the gradient, divergence, and curl in tensor form.				
Units	Course Content				Teaching Hrs
I	Differential operators: The concept of Gradient, Divergence, and Curl. Helmholtz decomposition.				9
II	Vector Integration: Line, surface, and volume integrals. Integral Theorems: Green's theorem in the plane, Gauss divergence theorems, Stokes' theorem, Green's formulas, and application of these theorems/formulas.				9
III	Curvilinear Coordinates: Curvilinear and orthogonal curvilinear coordinate systems and unit vectors. Arc lengths. Volume elements. Gradient, Divergence, and Curl in curvilinear coordinate systems. Some special orthogonal curvilinear coordinate systems.				9
IV	Tensor Analysis: Contra variant and covariant tensors, mixed tensors, coordinate transformation, and physical laws. Contraction, symmetric, and skew-symmetric tensors. metric tensor, length, and angle between two curves.				9
V	Christoffel symbols. Transformation laws of Christoffel symbols. Geodesics. Gradient, Divergence and Curl in tensor form, Derivation.				9
Texts / References	1. H. Guo: What are tensors exactly?, World Scienti_c, 2021. 2. D.C. Kay: Tensor Analysis, Schaum's Outline Series, McGraw Hill 1988. 3. R.S. Mishra: A Course in Tensors with Applications to Reimannian Geometry, Pothishala Pvt. Ltd, Allahabad. 4. M.R. Spiegel: Vector Analysis, Schaum's Outline Series, McGraw Hill, 1959.				
Learning Outcomes (not more than 100 words)	Upon completing this course, students can understand and apply the concepts of gradient, divergence, and curl and perform Helmholtz decomposition. They will evaluate line, surface, and volume integrals and apply integral theorems such as Green's, Gauss's, and Stokes' theorems. Students will master curvilinear coordinates, including orthogonal systems, and calculate arc lengths and volume elements. They will extend their understanding of tensor analysis by working with contravariant and covariant tensors, coordinate transformations, and the metric tensor. Additionally, students will understand Christoffel symbols and geodesics and express gradient, divergence, and curl in tensor form.				

Course Title		MATMJ43/ MATMN41: Differential Equations			
Credits & Hours of Teaching		Theory	Practical	Cumulative	
	Credits	4	0	4	
	Hours of Teaching (Total)	60	0	60	
Course Objectives (not more than 100 words)	This course aims to provide a thorough understanding of both ordinary differential equations (ODEs) and partial differential equations (PDEs). Students will explore linear differential equations of the first order, higher-degree equations solvable for x , y , z , singular solutions, and envelopes. The course covers linear differential equations with constant coefficients, homogeneous equations, and the method of variation of parameters. Additionally, students will study second-order linear differential equations with variable coefficients and series solutions. For PDEs, the course includes the formation and classification of linear first-order PDEs, Lagrange's method, Charpit's method for non-linear first-order PDEs, and higher-order linear PDEs with constant and variable coefficients.				
Units	Course Content				Teaching Hrs
I	Differential Equations: Linear differential equations of first order. First order higher degree equations solvable for x , y , z . Singular solution and envelopes.				15
II	Linear differential equations with constant coefficients, homogeneous linear differential equations.				15
III	Qualitative behaviour of solutions. Method of variation of parameters, linear differential equations of second order with variable coefficients, series solution.				15
IV	Partial Differential Equations: Formation of partial differential equations, Linear partial differential equations of first order and its classifications, Lagrange's method.				15
V	Non-linear PDE of first order: Charpit's method. Linear partial differential equation of second and higher order of homogeneous and non-homogeneous forms with constant coefficients, Linear partial differential equations reducible to equations with constant coefficients.				15
Texts / References	1. E.A. Coddington: An Introduction to Ordinary Differential Equations, Prentice Hall of India, 1961. 2. M. Braun: Differential equations and their applications. Springer-Verlag, New York, 1993. 3. B. Rai; D.P. Choudhary; H.I. Freedman: Ordinary Differential Equations, Narosa Publications, New Delhi, 2002. 4. S G Deo, V Raghavendra, Rasmita Kar, V Lakshmikantham Textbook of Ordinary Differential Equations, McGraw Hill Education (India) Private Limited, 2017. 5. G.F. Simmons, Differential equations with Applications and Historical notes, CRC Press. 6. I.N. Sneddon: Elements of Partial Differential Equations, McGraw-Hill Book Company, 1988.				
Learning Outcomes (not more than 100 words)	Upon completing this course, students can solve linear differential equations of the first order and higher-degree equations solvable for x , y , z . They will identify singular solutions and envelopes, handle linear differential equations with constant coefficients, and solve homogeneous equations. Students will understand the qualitative behavior of solutions and apply the method of variation of parameters. They will solve second-order linear differential equations with variable coefficients using series solutions. For PDEs, students will form and classify linear first-order PDEs, apply Lagrange's method, solve non-linear first-order PDEs using Charpit's method, and tackle higher-order linear PDEs with constant and variable coefficients.				

Course Title		MATMJ44: Mechanics			
Credits & Hours of Teaching		Theory	Practical	Cumulative	
	Credits	3	0	3	
	Hours of Teaching (Total)	45	0	45	
Course Objectives (not more than 100 words)	This course aims to provide a solid foundation in the mathematical analysis of forces and couples applied to particles and rigid bodies in static and dynamic states. Students will learn the concepts of vector addition and multiplication, differentiation of vector functions, and their applications in mechanics. The course covers the equilibrium of forces in 2D and 3D, the principle of virtual work, and Poinsot's central axis. It also explores the dynamics of a particle in two dimensions: central orbits, Kepler's laws, and constrained motion. Additionally, students will study the rigid body, moment and product of inertia, principle axes, and equi-momental systems.				
Units	Course Content				Teaching Hrs
I	Laws of vectors in two-dimension(2D): Application to mechanics, Concurrent forces in a plane, System of parallel forces, Coplanar forces, Theory of couples, Resultant Force and couple, Moment of a force about a point and about a line, Moment of a couple, Varignon's theorem.				9
II	Equilibrium of forces in 2D: Reduction and analytical conditions for the equilibrium of a system of forces acting on a particle and a rigid body, Different forms of the equilibrium conditions; Virtual displacement and virtual work, Principle of virtual work, Deduction of necessary and sufficient conditions for the equilibrium of a system of forces acting on a particle or different points of a rigid body. Application of Hooke's law related to virtual work.				9
III	Dynamics of a particle in 2D: Velocity, acceleration, angular velocity, linear momentum, angular momentum, relative velocity and acceleration, Expressions for velocity and acceleration for the cases of rectilinear and planar motions in Cartesian system, polar system and intrinsic system; Rectilinear motion: simple harmonic motion(SHM), Periodic motion, Amplitude, Time period, Frequency, Phase and epoch				9
IV	Central orbit and Kepler's laws of motion: Equation of central orbit in polar and pedal form, Motion under inverse square law, Particle moving along elliptic, parabolic, hyperbolic and other polar curves, Kepler's laws of planetary motion and related problems; Constrained motion on smooth vertical curves: Upward and downward motion along inner smooth surface of a vertical circle (SHM about the vertex), Upward and downward motion along inner smooth surface of a vertical cycloid (SHM about the vertex between two cusps), Time period of SHM and time taken to reach different heights of the cycloid.				9
V	Forces in 3D: Concept of axis of a couple, Reduction of forces and analytic conditions of equilibrium in 3D, Dyname and components of dyname, Poinsot's central axis; Definition of a rigid body, Moment of inertia(M.I.) and product of inertia(P.I.), Theorem of parallel axes, Moment of inertia and product of inertia of some symmetrical bodies: Rod, Rectangular plate, Circular ring and disc, Hollow and solid spheres, Principle axes and principal moments, M.I. and P.I. about an inclined lines.				9
Texts / References	1. J.L. Synge; B.A. Griffith: Principle of Mechanics, McGraw Hill, 1949. 2. F. Chorlton: Textbook of Dynamics, CBS Publishers, India, 2nd Edition, 2002. 3. S.L. Loney: An Elementary Treatise on Statics, G.K. Publishers, India, 2016. 4. S.L. Loney: An Elementary Treatise on Dynamics of Particle and of Rigid Body, AITS Publishers, India, 1998. 5. A.S. Ramsey: Dynamics (Part I & II), .CBS Publishers, India, 2002 6. R.C. Hibbelar: Engineering Mechanics-Statics and Dynamics, Prentice Hall Publishers, 13th Edition, 2012. 7. H. Goldstein: Classical mechanics, Addison-Wesley Series in Physics, Addison-Wesley Publishing Co., 2nd Edition, 1980.				
Learning Outcomes (not more than 100 words)	Upon completing this course, students will be able to understand and apply concepts of vectors to physical problems and analyze forces in equilibrium. They will comprehend the necessary conditions for equilibrium in two and three dimensions and apply the principle of virtual work. Students will understand particle dynamics, including velocity, acceleration, and motions in various coordinate systems. They will solve problems related to central orbits and Kepler's laws and recognize the dynamics of constrained motion. Students will also calculate moments and products of inertia and feel the concepts of principle axes, momental ellipse and ellipsoid, and equi-momental systems.				

Course Title		MATMJ51: Abstract Algebra			
Credits & Hours of Teaching		Theory	Practical	Cumulative	
	Credits	4	0	4	
	Hours of Teaching (Total)	60	0	60	
Course Objectives (not more than 100 words)	This course comprehensively introduces abstract algebra, covering fundamental concepts such as homomorphisms, isomorphisms, and group theory. Students will explore subgroups, group generation, and the center of a group, as well as Lagrange's theorem and its applications, including Euler's theorem and Fermat's theorem. The course will cover normal subgroups, quotient groups, and the fundamental theorems of homomorphism and isomorphism. Students will also study symmetric groups, permutations, and group actions. The course includes exploring rings, subrings, ideals, quotient rings, and polynomial rings, focusing on integral domains, division rings, and fields.				
Units	Course Content				Teaching Hrs
I	Homomorphism, Isomorphism. Subgroup, group generated by a subset, center of a group, coset, Lagrange theorem applications: Euler's theorem, Fermat and Wilson Theorem.				15
II	Product of groups, Normal subgroup, quotient group, Fundamental theorem of homomorphism, first and second isomorphism theorems, Commutator and abelianization.				15
III	Symmetric group, cycles, transpositions, even and odd permutations, Alternating group and normal subgroups of S_n .				15
IV	Group action: regular and conjugations, orbits, isotropy, class equation, conjugacy class, groups of order p^2 . Rings, definitions, and examples of subrings, rings of Gaussian integers, matrix rings, quaternions, Ideals, and quotient rings, Homomorphisms.				15
V	Integral domain, division rings, Communicative rings and fields. Field of fractions, Polynomial rings, division algorithm, Ideals of a polynomial ring over a field. Algebra of Reals, Complex numbers, Quaternions and Octonians.				15
Texts / References	1. I.N. Herstein: Topics in Algebra, Wiley Eastern Ltd, New Delhi, 1975. 2. M. Artin: Algebra, Prentice Hall, Inc., Englewood Cliffs, NJ, 1991. 3. J.B. Fraleigh: Algebra, Narosa, 1998. 4. R. Lal: Algebra vol 1, Springer, 2020.				
Learning Outcomes (not more than 100 words)	Upon completing this course, students can understand and apply the concepts of homomorphisms, isomorphisms, and subgroup structures, including Lagrange's theorem and its applications. They will analyze symmetric groups, permutations, and group actions and utilize concepts such as orbits and conjugacy classes. Students will explore ring theory, including the definitions and properties of subrings, ideals, and quotient rings. They will understand and work with integral domains, division rings, commutative rings, fields, and polynomial rings and apply the division algorithm and concepts related to the field of fractions.				

Course Title		MATMJ52: Metric Spaces			
Credits & Hours of Teaching		Theory	Practical	Cumulative	
	Credits	3	0	3	
	Hours of Teaching (Total)	45	0	45	
Course Objectives (not more than 100 words)	This course aims to understand metric spaces and functional analysis deeply. Students will explore the fundamental concepts of metric spaces, including neighborhoods, limit points, open and closed sets, and completeness. The course will cover compactness, Cantor's intersection theorem, and the construction of real numbers as the completion of rationals. Students will also study sequences and series of functions, focusing on pointwise and uniform convergence, Cauchy's criterion, and various tests for uniform convergence. Additionally, the course includes the study of uniform convergence about continuity, Riemann-Stieltjes integration, differentiation, and approximation theorems.				
Units	Course Content				Teaching Hrs
I	Definition of a Metric Space: Metric, examples, and basic properties. Open and Closed Sets: Definitions, examples, and properties within metric spaces. Neighborhoods and Convergence: Understanding neighborhoods, limit points, and convergent sequences. Topology induced by a metric. Interior, Closure, and Boundary: Definitions and their significance in metric spaces.				15
II	The subspace of a metric space. Completeness. Compactness: Heine-Borel Theorem. Compactness and sequential compactness. Connectedness: Connected and disconnected sets. Path-connected spaces.				15
III	Cantor's intersection theorem. Construction of real numbers as the completion of the incomplete metric space of rationals. Dense subsets. Separable metric spaces. Continuous functions. Uniform continuity, Isometry, and homeomorphism. Equivalent metrics.				15
Texts / References	1. E.T. Copson: Metric Spaces, Cambridge University Press, 1968. 2. S. Kumaresan: Topology of Metric Spaces, Narosa Publishing House, New Delhi, 2005. 3. W. Rudin: Principles of Mathematical Analysis, McGraw-Hill, 1976. 4. B.K. Tyagi: First Course in Metric Spaces, Cambridge University Press, 2010. 5. Irving Kaplansky, Set Theory and Metric Spaces, AMS Chelsea Publishing, 1972. 6. Mícheál Ó Searcóid, Metric Spaces, Springer, 2006. 7. Athanase Papadopoulos, Metric Spaces, Convexity and Nonpositive Curvature: Second Edition, European Mathematical Society, 2014.				
Learning Outcomes (not more than 100 words)	Upon completing this course, students will be able to understand and apply concepts related to metric spaces, including neighborhoods, limit points, and the notions of open and closed sets. They will analyze completeness, compactness, and Cantor's intersection theorem and construct real numbers as the completion of rationals. Students will explore dense subsets, separable metric spaces, and continuous functions. They will analyze sequences and series of functions for pointwise and uniform convergence, apply Cauchy's criterion, and utilize convergence tests. Students will understand the relationship between uniform convergence, continuity, and Riemann-Stieltjes integration and apply approximation theorems and equicontinuity concepts.				

Course Title		MATMJ53: Analytic Geometry			
Credits & Hours of Teaching		Theory	Practical	Cumulative	
	Credits	4	0	4	
	Hours of Teaching (Total)	60	0	60	
Course Objectives (not more than 100 words)	This course offers a detailed exploration of polar coordinates and their applications and the study of geometric entities in three-dimensional space. Students will learn to work with polar equations for lines, circles, and conics and understand the equations for chords, tangents, and normals. The course covers the analysis of planes and straight lines in $\mathbb{R}^3$ using vector methods and the properties of spheres, including plane sections, circles, and tangent planes. Students will also study cones and cylinders, including their equations and conditions of tangency, and explore conoids, including central conoids and paraboloids, emphasizing tangent planes, normals, and the classification of quadrics.				
Units	Course Content				Teaching Hrs
I	Polar coordinators, Polar Equation: Distance between two points. Polar equation of a straightline. Polar equation of a circle. Polar equation of a conic. Equations of chord, tangent, normal, chord of contact of a conic.				12
II	Plane and straight line: Plane and straight line in $\mathbb{R}^3$ by using vector method.				12
III	Sphere: Equation of a sphere. Plane section of sphere. Equations of circle. Sphere passing through a given circle. Equation of tangent plane. Angle of intersection of two sphere.				12
IV	Cone and Cylinder: Cone with vertex as origin. Condition for the general equation of second degree to represent a cone. Equation of Right circular cone. Equation of tangent plane to a cone. Condition of tangency. Reciprocal cone. Enveloping cylinder, Right circular cylinder.				12
V	Conicoids: Central conicoids. Paraboloids. Equation of tangent plane and normal to a conicoid. Condition of tangency. Cone through six normals. Plane sections of conicoids. Enveloping cone and enveloping cylinder. Classification of quadrics.				12
Texts / References	1. R.J.T. Bell: An Elementary Treatise on Co-ordinate geometry of three dimensions, Macmillan India Ltd., New Delhi, 1994. 2. M.M. Tripathi: Coordinate Geometry: Polar Coordinates Approach, Narosa Publishing House, New Delhi, 2005. 3. M.M. Tripathi: A Course on Geometry in Euclidean 3-space, 4. P. Balasubramanyam; K.G. Subramanyam; G.R. Venkatraman: Coordinate geometry of two and three dimensions, Tata McGraw-Hill, New Delhi, 1994.				
Learning Outcomes (not more than 100 words)	Upon completing this course, students can derive and analyze polar equations for lines, circles, and conics and understand the equations for associated geometric features such as chords and tangents. Using vector methods, they will work with planes and straight lines in three-dimensional space. Students will understand and apply the equations for spheres, including plane sections and tangent planes, and analyze intersections of spheres. They will study cones and cylinders, including conditions for tangency and equations of enveloping surfaces. Additionally, students will explore conicoids, including paraboloids, classify quadrics, and understand tangent planes, normals, and other geometric properties.				

Course Title		MATMJ54: Numerical Analysis			
Credits & Hours of Teaching		Theory	Practical	Cumulative	
	Credits	3	1	4	
	Hours of Teaching (Total)	45	15	60	
Course Objectives (not more than 100 words)	This course focuses on numerical methods and their applications in solving mathematical problems. Students will learn techniques for computing errors and solving nonlinear equations using Bisection, Regula-Falsi, Secant, and Newton-Raphson methods, along with fixed-point iteration and the convergence rate. The course covers the solution of polynomial roots using the Birge-Vieta method, linear systems using Gauss elimination and iterative methods, and eigenvalue computation through the Power method. Students will also explore finite differences and interpolation techniques, including Newton's forward and backward methods, Lagrange interpolation, and divided difference interpolation. Key numerical integration methods, including the Trapezoidal, Simpson, Boole, and Weddle's rules, will be covered. The course concludes with methods for solving ordinary differential equations, including Euler's method, Modified Euler's method, Runge-Kutta methods, and Predictor-Corrector methods.				
Units	Course Content				Teaching Hrs
I	Errors and their computations; Numerical solutions of nonlinear equations: Bisection method, Regula-Falsi, Secant method, Newton-Raphson, fixed point iteration, Rate of convergence of iterative methods;				9
II	Roots of Polynomials: Birge-Vieta method; System of linear equations: Gauss elimination method, Gauss-Jordan method, Jacobi iterative method, Gauss-Seidal iterative method; Eigen value computation: Power method; Finite differences;				9
III	Interpolation: Newton's forward and backward interpolation, Lagrange's interpolation, Newton's divided difference interpolation; Numerical differentiation;				9
IV	Numerical integration: Trapezoidal rule, Simpson's one-third rule, Simpson's three-eighth rule, Boole's rule and Weddle's rule; Errors in quadrature formulae;				9
V	Numerical solution to ordinary differential equations of first order: Euler's method, Modified Euler's method, Runge-Kutta second and fourth order, Implicit Runge-Kutta second order; Predictor Corrector methods: Milne Simpson method, Adams-Bashforth method.				9
Texts / References	1. M.K. Jain; S.R.K. Iyengar; R.K. Jain: Numerical Methods for Scientific and Engineering Computation, New Age International, New Delhi, 1984. 2. C.F. Gerald; P.O. Wheatley: Applied Numerical Analysis, Pearson Education, 1970. 3. S.D. Conte; C. de Boor: Elementary Numerical Analysis, McGraw-Hill, 1972. 4. C.E. Froberg: Introduction to Numerical Analysis, Addison-Wesley, 1965. 5. M.J. Maron, Numerical Analysis A Practical Approach, Macmillan Publishing Company Inc., New York, 1982. 6. S.S. Sastry: Introductory Methods of Numerical Analysis, PHI Learning Private Limited, New Delhi, 1979.				
Learning Outcomes (not more than 100 words)	Upon completing this course, students will be proficient in various numerical techniques for solving mathematical problems. They can compute and assess errors and apply methods for solving nonlinear equations, including iterative techniques and convergence analysis. Students will gain expertise in solving systems of linear equations and computing eigenvalues. They will understand and apply interpolation techniques and numerical differentiation and accurately perform numerical integration using multiple quadrature rules. Additionally, students will be skilled in solving ordinary differential equations using various methods, including Euler's, Runge-Kutta, and Predictor-Corrector methods.				

Course Title	MATMV51: Optimization Techniques			
Credits & Hours of Teaching		Theory	Practical	Cumulative
	Credits	4	0	4
	Hours of Teaching (Total)	60	0	60
Course Objectives (not more than 100 words)	This course aims to equip students with fundamental and advanced techniques in optimization, focusing on linear programming (LP), duality, and both constrained and unconstrained optimization. Students will learn model formulation, graphical methods, and the Simplex Method for LP problems while exploring duality and specialized algorithms like the transportation problem. Key methods such as steepest descent and conjugate gradients will be covered for unconstrained optimization alongside necessary and sufficient optimality conditions. In constrained optimization, the focus will be on KKT conditions and Lagrangian functions, with practical applications across industries			
Units	Course Content			Teaching Hrs
I	Real life Applications and Model Formulation of Linear Programming problems, Graphical Methods, convex set, extreme points, Canonical and Standard Form, Slack, Surplus and artificial Variables, basic feasible solution, The Simplex Method, Two-Phase Method, Big-M Method.			15
II	Duality in Linear Programming, dual simplex method, Transportation Problem, North-West Corner Method, Least Cost Method, Vogel's Approximation Method, MODI method, assignment problem, Hungarian Method.			15
III	Unconstrained Optimization, local and global minimizers, first and second-order necessary optimality conditions, second-order sufficient optimality condition, steepest descent method, the conjugate gradient method.			15
IV	Constrained Optimization, feasible directions, descent directions, Lagrangian function, Fritz-John optimality conditions, constraint qualifications, Karush-Kuhn-Tucker optimality conditions, convex functions and properties, sufficient optimality conditions, applications.			15
Texts / References	1. D.G. Luenberger; Y. Ye: Linear and Nonlinear Programming, Springer, 2008. 2. DG. Luenberger: Optimization by Vector Space Methods, Wiley-Interscience, 1997. 3. S.S. Rao: Engineering Optimization: Theory and Practice, Newage Publishers, 2013.			
Learning Outcomes (not more than 100 words)	By the end of the course, students will: Formulate and solve linear programming problems using graphical and simplex methods. Understand the concepts of duality and sensitivity analysis in LP. Apply transportation and assignment problem techniques. Utilize optimization algorithms like the steepest descent and conjugate gradient methods for unconstrained problems. Derive and interpret optimality conditions (KKT, Lagrangian) in constrained problems. Analyze convexity and apply sufficient conditions to optimization models in real-world applications.			

Course Title		MATMJ61: Measure and Integration			
Credits & Hours of Teaching		Theory	Practical	Cumulative	
	Credits	4	0	4	
	Hours of Teaching (Total)	60	0	60	
Course Objectives (not more than 100 words)	This course provides an in-depth exploration of measure theory and integration, focusing on developing and applying measures and outer measures. Students will study the extension and uniqueness of measures and the completion of measures with a detailed look at Lebesgue outer measure and its properties. The course covers measurable sets and functions, Borel and Lebesgue measurability, and the integration of non-negative functions, including convergence theorems and comparisons between Riemann and Lebesgue integrals. Students will also explore signed measures, the Hahn and Jordan decomposition theorems, the Radon-Nikodym theorem, and Lebesgue-Stieltjes integrals, with applications to product measures, Fubini's theorem, and measures on locally compact spaces.				
Units	Course Content				Teaching Hrs
I	Measures and outer measures. Measure induced by an outer measure, Extension of a measure. Uniqueness of Extension, Completion of a measure. Lebesgue outer measure. Measurable sets.				12
II	Non-Lebesgue measurable sets. Regularity. Measurable functions. Borel and Lebesgue measurability. Integration of non-negative functions. The general integral. Convergence theorems. Mode in Convergence, Riemann and Lebesgue integrals.				12
III	Signed measure. Hahn and Jordan decomposition theorems. Absolutely continuous and singular measures.				12
IV	Radon-Nikodym theorem. Lebesgue decomposition. Riesz representation theorem. Extension theorem (Carathéodory). Lebesgue-Stieltjes integral.				12
V	Product measures. Fubini's theorem. Baire sets. Baire measure. Continuous functions with compact support. Regularity of measures on locally compact spaces. Integration of continuous functions with compact support. Riesz-Markoff theorem.				12
Texts / References	1. H.L. Royden: Real Analysis, 4th Edition, Macmillan, 1993. 2. P.R. Halmos: Measure Theory, Van Nostrand, 1950. 3. G. de Barra: Measure Theory and Integration, Wiley Eastern, 1981. 4. E. Hewitt; K. Stromberg: Real and Abstract Analysis, Springer, 1969. 5. R. G. Bartle: The Elements of Integration, John Wiley, 1966. 6. I.K. Rana: An Introduction to Measure and Integration, Narosa Publishing House, 1997.				
Learning Outcomes (not more than 100 words)	Upon completing this course, students can understand and apply concepts related to measures and outer measures, including their extensions, uniqueness, and completion. They will identify and work with measurable sets and functions and understand the differences between Borel and Lebesgue measurability. Students will be proficient in integrating non-negative functions, using convergence theorems, and comparing Riemann and Lebesgue integrals. They will also analyze signed measures, apply the Hahn and Jordan decomposition theorems, and use the Radon-Nikodym theorem. Additionally, students can work with the Lebesgue-Stieltjes integral, product measures, Fubini's theorem, and measures on locally compact spaces.				

Course Title		MATMJ62: Complex Analysis			
Credits & Hours of Teaching		Theory	Practical	Cumulative	
	Credits	4	0	4	
	Hours of Teaching (Total)	60	0	60	
Course Objectives (not more than 100 words)	This course delves into the theory and applications of complex analysis, exploring the fundamental concepts of the complex plane, including open sets, domains, and regions. Students will study stereographic projections, linear transformations, and Möbius transformations, focusing on their geometric properties and conformal mappings. Key topics include the limits, continuity, differentiability, and analyticity of complex functions governed by the Cauchy-Riemann equations. The course covers complex integration, including line integrals, Cauchy's theorems, and the Cauchy integral formula. Students will also explore series expansions, including Taylor and Laurent series, and investigate singularities, branch points, and multi-valued functions.				
Units	Course Content				Teaching Hrs
I	The complex plane and open set, domain, and region in a complex plane. Stereographic projection. Linear transformations, the transformation $w = 1/z$, Möbius transformations and its geometric properties, Conformal mappings.				12
II	Schwarz' theorem, Riemann mapping theorem and its applications. Functions of complex variables and their mapping properties. Elementary functions: the exponential, logarithmic, and trigonometric functions. Limits, continuity, differentiability, and analyticity of functions of a complex variable. The Cauchy-Riemann equations and sufficient conditions for differentiability and analyticity. Harmonic functions.				12
III	Complex integration: Line integration, Green's theorem, antiderivative theorem, Cauchy's theorem (weak form), Cauchy-Goursat's theorem, Cauchy's integral formula, Cauchy's inequality, Liouville theorem, fundamental theorem of algebra, maximum modulus theorem.				12
IV	Sequences, series and their convergence, power series, radius of convergence, Taylor and Laurent series.				12
V	Integration and differentiation of power series, Absolute and uniform convergence of power series. Branch point, branch cut, branches of a multi-valued function, analyticity of the branches of $\text{Log } z, z^a$. Singularities and their classification, Weierstrass-Casorati's theorem.				12
Texts / References	1. T. Apostol, Mathematical Analysis, Addison-Wesley, Reading, MA, 1967. 2. L.V. Ahlfors, Complex analysis, McGraw-Hill International Editions, New Delhi, 1979. 3. J.E. Brown, R.V. Churchill, Complex Variables and Applications, McGraw-Hill, 2004. 4. J.B. Conway, Functions of one complex variable, Springer-Verlag, 1978. 5. T.W. Gamelin, Complex analysis, Springer, 2001. 6. R.E. Greene, S.G. Krantz, Function Theory of One Complex Variable, John Wiley and Sons, Inc., 1997. 7. S. Ponnusamy, Foundations of Functional Analysis, Narosa Publishing House, 2005. 8. S. Ponnusamy, H. Silverman, Complex variables with applications, Birkhäuser, 2006. 9. W. Rudin, Real and Complex Analysis, Third ed., McGraw-Hill, New York, 1987.				
Learning Outcomes (not more than 100 words)	Upon completing this course, students can understand and apply concepts related to the complex plane, including open sets, domains, and stereographic projections. They will analyze linear and Möbius transformations, their geometric properties, and conformal mappings. Students will master the limits, continuity, differentiability, and analyticity of complex functions using the Cauchy-Riemann equations. They will be proficient in complex integration, including applying Cauchy's theorems and integral formulas. Students will also be able to work with power series, determine their radius of convergence, and understand branch points and singularities.				

Course Title		MATMJ63: Differential Geometry			
Credits & Hours of Teaching		Theory	Practical	Cumulative	
	Credits	4	0	4	
	Hours of Teaching (Total)	60	0	60	
Course Objectives (not more than 100 words)	This course thoroughly examines differential geometry, focusing on curves and surfaces in $\mathbb{R}^2$ and $\mathbb{R}^3$. Students will explore fundamental concepts such as arc length, curvature, and the Frenet-Serret apparatus for curves, including the rotation index and various theorems related to curves. For surfaces, the course covers essential topics such as the First and Second Fundamental Forms, normal curvature, geodesic curvature, and key theorems, including Gauss's Theorema Egregium and the Gauss-Bonnet Formula. Emphasis is placed on understanding the intrinsic and extrinsic properties of surfaces, isometries, and geodesics.				
Units	Course Content				Teaching Hrs
I	Curves in $\mathbb{R}^2$ and $\mathbb{R}^3$: Basic Definitions and Examples. Arc Length. Curvature and the Frenet-Serret Apparatus. The Fundamental Existence and Uniqueness Theorem for Curves. Bishop frames.				12
II	Non-Unit Speed Curves. The Rotation Index of a Plane Curve, Convex Curves, The Isoperimetric Inequality, Mukhopadhyay Theorem (The Four-Vertex Theorem), Fenchel's Theorem, The Fary-Milnor Theorem.				12
III	Surfaces in $\mathbb{R}^3$: Basic Definitions and Examples. The First Fundamental Form. Arc length of curves on surfaces. Normal curvature. Geodesic curvature. Gauss and Weingarten Formulas.				12
IV	Geodesics, Parallel Vector Fields Along a Curve, and Parallelism. The Second Fundamental Form and the Weingarten Map, Principal, Gaussian, and Mean Curvatures. Casorati curvature.				12
V	Isometries of surfaces, Gauss's Theorema Egregium, The Fundamental Theorem of Surfaces, Surfaces of Constant Gaussian Curvature. Exponential map, Gauss Lemma, Geodesic Coordinates. The Gauss-Bonnet Formula and the Gauss-Bonnet Theorem (description only).				12
Texts / References	1. Christian Bär, Elementary Differential Geometry, Cambridge University Press, 2010. 2. M. P. do Carmo, Differential geometry of curves and surfaces, Prentice Hall 1976. 3. A. Gray, Differential Geometry of Curves and Surfaces, CRC Press, 1998. 4. R. S. Millman and G. D. Parkar, Elements of Differential Geometry, Prentice Hall 1977. 5. S. Montiel and A. Ros, Curves and Surfaces, American Mathematical Society, 2005. 6. S. Mukhopadhyay, Parametric Coefficients in the Differential Geometry of Curves, Calcutta University Press, 1910. 7. B. O'Neill, Elementary Differential Geometry, Elsevier 2006. 8. John Oprea, Differential Geometry and its applications, Prentice Hall 1997. 9. A. Pressley, Elementary Differential Geometry, Springer 2010. 10. John A. Thorpe, Elementary Topics in Differential Geometry, Springer, 1979. 11. V. A. Toponogov, Differential geometry of curves and surfaces – A concise guide, Birkhäuser, 2006.				
Learning Outcomes (not more than 100 words)	Upon completing this course, students will understand differential geometry deeply, focusing on studying curves and surfaces in $\mathbb{R}^2$ and $\mathbb{R}^3$. They can compute and interpret key geometric properties such as arc length, curvature, and the Frenet-Serret apparatus for curves. Students will also master the rotation index of plane curves and apply various theorems, including Mukhopadhyay's Four-Vertex Theorem, Fenchel's Theorem, and the Fary-Milnor Theorem. In studying surfaces, they will learn to analyze the intrinsic properties using the First and Second Fundamental Forms, normal and geodesic curvatures, and apply Gauss and Weingarten formulas. Additionally, students will understand isometries, Gauss's Theorema Egregium, and the Gauss-Bonnet Formula, equipping them with the tools to explore surface curvature and geometric properties in depth.				

Course Title		MATMJ64: Number Theory			
Credits & Hours of Teaching		Theory	Practical	Cumulative	
	Credits	3	0	3	
	Hours of Teaching (Total)	45	0	45	
Course Objectives (not more than 100 words)	This course aims to introduce fundamental concepts in number theory, including divisibility, prime numbers, GCD, LCM, and unique factorization. Students will explore Euclidean algorithms, congruences, and modular arithmetic. The course will cover solving Diophantine equations, quadratic residues, Gaussian integers, and prime number distribution. Advanced topics include the Riemann Zeta function, prime number theorems, and applications in cryptography and coding.				
Units	Course Content				Teaching Hrs
I	Divisibility, prime numbers, greatest common divisor (GCD), least common multiple (LCM). Fundamental Theorem of Arithmetic: Unique factorization in integers. Euclidean Algorithm: Computing GCD, extended Euclidean algorithm.				9
II	Congruences and Modular Arithmetic: Definitions, properties. Linear Congruences: Solvability, algorithms. Chinese Remainder Theorem: Statement and applications. Diophantine Equations Linear Diophantine Equations: Solving equations of the form $ax + by = c$.				9
III	Fermat's Last Theorem: Historical context, basic proof ideas. Pell's Equation: Methods of solving, applications. Quadratic Residues and Quadratic Reciprocity Quadratic				9
IV	Residues: Definitions, properties. Quadratic Reciprocity: Statements, applications. Gaussian Integers: Definition, properties. Prime Numbers Prime Number Distribution: Sieve methods, prime number theorem. Dirichlet's Theorem: Arithmetic progressions, primes in progressions.				9
V	Goldbach's Conjecture: Statement, historical context. Analytic Number Theory Riemann Zeta Function: Definition, analytic continuation. Prime Number Theorems: Statements and elementary proofs. Distribution of Primes: Asymptotic behavior, twin primes conjecture. RSA algorithm, applications of number theory in cryptography. Error-correcting codes, applications of number theory in coding.				9
Texts / References	- David M. Burton, Elementary Number Theory, Wm. C. Brown Publishers, Dubuque, Iowa 1989. K. Ireland, and M. Rosen, A Classical Introduction to Modern Number Theory, GTM Vol. 84, Springer-Verlag, 1972. G. A. Jones, and J. M. Jones, Elementary Number Theory, Springer-Verlag, 1998. W. Sierpinski, Elementary Theory of Numbers, North-Holland, Ireland, 1988. - Niven, S.H. Zuckerman, and L.H. Montgomery, An Introduction to the Theory of Numbers, John Wiley, 1991. H. B. Mann, Addition Theorems, Krieger, 1976. - Melvyn B. Nathanson, Additive Number Theory: Inverse Problems and the Geometry of Subsets, Springer-Verlag, 1996.				
Learning Outcomes (not more than 100 words)	Upon completion, students will be proficient in the foundational principles of number theory and their applications. They will understand and apply the Euclidean algorithm, solve linear Diophantine equations, and work with modular arithmetic. Students will also grasp advanced concepts such as the distribution of primes, the RSA algorithm, and using number theory in cryptography and error-correcting codes.				

Course Title		MATMJ65: Linear Programming and Applications			
Credits & Hours of Teaching		Theory	Practical	Cumulative	
	Credits	3	0	3	
	Hours of Teaching (Total)	45	0	45	
Course Objectives (not more than 100 words)	This course provides a comprehensive introduction to linear programming and its applications. Students will learn to formulate and solve linear programming problems using graphical methods, the Simplex Method, and its variants, including the Two-Phase and Big-M Methods. The course covers key concepts in linear programming theory, such as canonical and standard forms, slack, surplus variables, basic feasible solutions, and issues like degeneracy and unbounded solutions. Students will explore duality in linear programming, including the formulation and solution of dual problems and the dual simplex method. The course also addresses specialized problems, such as transportation and assignment problems, with the North-West Corner Method, Least Cost Method, Vogel's Approximation Method, and the Hungarian Method.				
Units	Course Content				Teaching Hrs
I	Linear Programming: Applications and Model Formulation, Graphical Solution Methods, The Simplex Method, Two-Phase Method, Big-M Method, Some Complications and Their Resolution, Different types of Linear Programming Solutions.				9
II	Theory of Simplex Method: Canonical and Standard Form of LP Problem, Slack, Surplus and artificial Variables, basic solution, degenerate solution, Reduction of Feasible Solution to a Basic Feasible Solution, Improving a Basic Feasible Solution, Alternative Optimal Solutions, Unbounded Solution, Optimality Condition, Unrestricted Variables, Degeneracy and its Resolution.				9
III	Duality in Linear Programming, Formulation of Dual Linear Programming Problem, Standard Results on Duality, dual simplex method, revised simplex method, Integer Linear Programming.				9
IV	Transportation Problem: Mathematical Model of Transportation Problem, North-West Corner Method, Least Cost Method, Vogel's Approximation Method, Test for Optimality by MODI method, Variations in Transportation Problem, Maximization Transportation Problem.				9
V	Assignment Problem: Mathematical Models of Assignment Problem, Hungarian Method for Solving Assignment Problem.				9
Texts / References	1. Ferris, M. C., Mangasarian, O. L., & Wright, S. J., Linear programming with MATLAB. Society for Industrial and Applied Mathematics., 2007. 2. Kwon, R. H., Introduction to linear optimization and extensions with MATLAB. CRC Press, 2013. 3. Ploskas, N., and Samaras, N., Linear programming using MATLAB (Vol. 127). Switzerland: Springer, 2017. 4. Mishra, S. K., and Ram, B., Introduction to linear programming with MATLAB. CRC Press, 2017.				
Learning Outcomes (not more than 100 words)	Upon completing this course, students can effectively formulate and solve linear programming problems using various methods, including graphical solutions and the Simplex Method. They will understand and apply the theory behind canonical and standard forms, slack and surplus variables, and handle issues such as degeneracy and unbounded solutions. Students will be adept at formulating and solving dual linear programming problems, utilizing dual and revised simplex methods. Additionally, they will solve transportation and assignment problems using specific methods and criteria for optimality.				

Course Title		MATMJ66: Special Theory of Relativity			
Credits &Hours of Teaching		Theory	Practical	Cumulative	
	Credits	3	0	3	
	Hours of Teaching (Total)	45	0	45	
Course Objectives (not more than 100 words)	The course aims to deepen students' understanding of the foundations and implications of the special theory of relativity. It will review Newtonian mechanics, introduce concepts such as inertial frames, the Michelson-Morley experiment, and Lorentz transformations, and explore the relativistic treatment of space and time. The course also covers relativistic kinematics, the geometric representation of space-time, and the relationship between mass and energy. Additionally, it examines the relativistic formulations of mechanics and electromagnetism, including Maxwell's equations and their transformations.				
Units	Course Content				Teaching Hrs
I	Review of Newtonian mechanics: Inertial frames. Speed of light and Gallilean relativity. Michelson-Morley experiment. Lorentz-Fitzgerold contraction hypothesis. Relative character of space and time. Postulates of special theory of relativity. Lorentz transformation equations and its geometrical interpretation. Group properties of Lorentz transformations.				9
II	Relativistic kinematics: Composition of parallel velocities. Length contraction. Time dilation. Transformation equations for components of velocity and acceleration of a particle and Lorentz contraction factor.				9
III	Geometrical representation of space-time: Four dimensional Minkowskian space-time of special relativity. Time-like, light-like and space-like intervals. Null cone, Proper time. World line of a particle. Four vectors and tensors in Minkowskian space-time.				9
IV	Relativistic mechanics – Variation of mass with velocity. Equivalence of mass and energy. Transformation equations for mass momentum and energy. Energy-momentum four vector. Relativistic force and Transformation equations for its components. Relativistic Lagrangian and Hamiltonian. Relativistic equations of motion of a particle.				9
V	Energy momentum tensor of a continuous material distribution. Electromagnetism - Maxwell's equations in vacuo. Transformation equations for the densities of electric charge and current. Propagation of electric and magnetic field strengths. Transformation equations for electromagnetic four potential vector. Transformation equations for electric and magnetic field strengths. Gauge transformation. Lorentz invariance of Maxwell's equations. Maxwell's equations in tensor form. Lorentz force on a charged particle. Energy momentum tensor of an electromagnetic field.				9
Texts / References	1. C. Moller, The Theory of Relativity, Oxford Clarendon Press, 1952. 2. P. G. Bergmann, Introduction to the Theory of Relativity, Prentice Hall of India, 1969. 3. J. L. Anderson, Principles of Relativity Physics, Academic Press, 1967. 4. W. Rindler, Essential Relativity, Van Nostrand Reinhold Company, 1969. 5. V. A. Ugarov, Special Theory of Relativity, Mir Publishers, 1979. 6. R. Resnick, Introduction to Special Relativity, Wiley Eastern Pvt. Ltd. 1972. 7. J. L. Synge, Relativity: The Special Theory, North-Holland Publishing Company, 1956. 8. W. G. Dixon, Special Relativity: The Foundation of Macroscopic Physics, Camb. Uni. Press, 1982.				
Learning Outcomes (not more than 100 words)	By the end of the course, students will be able to explain the key concepts of special relativity, including time dilation, length contraction, and the relativistic transformation of velocities and accelerations. They will understand the geometric representation of space-time and the role of four vectors and tensors. Students can also derive and apply the transformation equations for physical quantities in electromagnetism and mechanics, comprehend the energy-momentum tensor, and appreciate the Lorentz invariance of physical laws.				

Course Title		MATMJ67: Space dynamics			
Credits & Hours of Teaching		Theory	Practical	Cumulative	
	Credits	3	0	3	
	Hours of Teaching (Total)	45	0	45	
Course Objectives (not more than 100 words)	The course comprehensively introduces celestial mechanics and its relevance in aerospace engineering. It covers Kepler's laws, orbital mechanics, the two-body problem, and orbital maneuvers and perturbation theory. The curriculum extends to special cases like the restricted three-body problem, attitude dynamics and control, spacecraft dynamics, and mission design. It also explores advanced topics and emerging trends in space dynamics, complemented by practical laboratory sessions using simulation tools.				
Units	Course Content				Teaching Hrs
I	Introduction to Celestial Mechanics Overview of space dynamics and its importance in aerospace engineering Kepler's laws of planetary motion Two-body problem: orbit determination, orbital elements. Orbital Mechanics Keplerian orbits: elliptical, circular, parabolic, hyperbolic Orbital parameters and their significance Orbital maneuvers: Hohmann transfer, inclination change, rendezvous maneuvers.				9
II	Perturbation Theory Effects of perturbations: gravitational, atmospheric, solar radiation pressure Perturbation analysis: secular effects, periodic effects Methods of orbit determination considering perturbations.				9
III	Orbital Mechanics in Special Cases Restricted three-body problem: Lagrange points, stability regions Orbital resonance and its implications.				9
IV	Attitude Dynamics and Control Attitude representation: Euler angles, quaternion Attitude dynamics: equations of motion, torque analysis Attitude control systems: control strategies, stabilization techniques. Spacecraft Dynamics Spacecraft kinematics and dynamics Equations of motion for a rigid body in space Spacecraft stability and control: stability criteria, control actuators.				9
V	Space Mission Design Mission requirements and objectives Mission analysis: trajectory optimization, launch window determination Case studies: interplanetary missions, satellite constellations . Advanced Topics and Emerging Trends Advanced topics in space dynamics: chaotic motion, libration points Emerging trends in space dynamics research Laboratory Sessions and Projects Simulation of orbital mechanics using software tools (e.g., MATLAB, STK)				9
Texts / References	1. H.D. Curtis: "Orbital Mechanics for Engineering Students." Butterworth-Heinemann, 3rd edition, 2013. 2. D.A. Vallado: "Fundamentals of Astrodynamics and Applications." Microcosm Press, 4th edition, 2013. 3. J.E. Prussing; and Bruce A. Conway. "Orbital Mechanics." Oxford University Press, 1993.				
Learning Outcomes (not more than 100 words)	By the end of the course, students will understand the fundamental principles of celestial mechanics, including the determination of orbits and orbital elements. They can analyze and execute orbital maneuvers, account for perturbations, and understand attitude dynamics and control strategies. Students will develop the skills to design space missions, including trajectory optimization and stability analysis. They will also gain practical experience in simulating orbital mechanics using software tools and be familiar with advanced concepts and current research trends in space dynamics.				

Course Title		MATMJ68: Discrete Mathematics			
Credits & Hours of Teaching		Theory	Practical	Cumulative	
	Credits	3	0	3	
	Hours of Teaching (Total)	45	0	45	
Course Objectives (not more than 100 words)	The course aims to introduce students to key concepts in discrete mathematics, focusing on counting techniques, lattice theory, and Boolean algebra. It covers the structure and properties of lattices, including duality and distributive properties, and explores Boolean functions and their applications to switching circuits. Additionally, the course introduces graph theory, including studying different types of graphs, paths, circuits, and matrix representations, focusing on Eulerian paths, planar graphs, and trees.				
Units	Course Content				Teaching Hrs
I	Counting techniques. Lattices as partially ordered sets and as algebraic systems.				9
II	Duality, Distributive, complemented and complete lattices.				9
III	Boolean algebras and their basic properties. Boolean functions and expressions. Application of Boolean algebra to switching circuits (using AND, OR and NOT gates).				9
IV	Graphs and Planar Graphs: Graphs, Multi-graphs, Weighted Graphs, Directed graphs. Paths and circuits.				9
V	Matrix representation of graphs. Eulerian paths and circuits. Planar graphs. Euler's formula. Trees and spanning trees.				9
Texts / References	1. J.P. Tremblay; R. Manohar: Discrete Mathematical Structures with Applications to Computer Science, Tata McGraw Hill, New Delhi, 2001. 2. C.L. Liu: Elements of Discrete Mathematics, 2nd Edition, Tata McGraw-Hill, International Edition, 1986. 3. K.H. Rosen: Discrete Mathematics and its Applications, 8th Edition, Tata McGraw-Hill, 2021. 4. S. Wiitala: Discrete Mathematics: A Unified Approach, McGraw-Hill Book Co., 1987. 5. N. Deo: Graph Theory with Applications to Engineering and Computer Science, Prentice-Hall of India. 6. V.K. Balakrishnan: Introductory Discrete Mathematics, Dover, 1996.				
Learning Outcomes (not more than 100 words)	By the end of the course, students will be able to apply counting techniques and understand the fundamental concepts of lattice theory, including Boolean algebra and their applications in digital circuits. They will also gain proficiency in graph theory, classifying and analyzing different types of graphs, identifying Eulerian paths and circuits, applying Euler's formula to planar graphs, and understanding the properties and applications of trees and spanning trees.				

Course Title		MATMJ69: Mathematical Modeling			
Credits & Hours of Teaching		Theory	Practical	Cumulative	
	Credits	3	0	3	
	Hours of Teaching (Total)	45	0	45	
Course Objectives (not more than 100 words)	The course aims to provide an understanding of differential equations and their role in modeling dynamic systems. It covers the concepts of fixed points, stability analysis, and bifurcations. Students will learn the principles of mathematical modeling, including the system approach, formulation, and analysis of models. The course includes discrete and continuous-time models, such as cell growth, logistic, and prey-predator models, and introductory epidemic models like SIS and SIR.				
Units	Course Content				Teaching Hrs
I	System of differential equations; Fixed points and their stability; Basic idea of bifurcations.				9
II	Introduction to modeling: Elementary mathematical models and General modeling ideas.				9
III	Role of mathematics in problem-solving; Concepts of mathematical modeling; system approach; formulation, Analysis of models.				9
IV	Discrete and Continuous-time models: cell growth model, logistic model, prey-predator model.				9
V	Introduction to infectious diseases: Simple epidemic models such as SIS, SIR, etc.				9
Texts / References	1. J.N. Kapur: Mathematical Model in Biology and Medicine, East-West Press Pvt. Ltd. 2. S.H. Strogatz: Nonlinear dynamics and chaos with student solutions manual: With applications to physics, biology, chemistry, and engineering. CRC Press; 2018. 3. C.L. Dyne: Principles of Mathematical Modeling, Academic Press, 2004. 4. D.N.P. Murthy; N.W. Page; E. Y. Rodin: Mathematical Modelling: A Tool for Problem Solving in Engineering, Physical, Biological and Social Sciences, Pergamon 1990.				
Learning Outcomes (not more than 100 words)	By the end of the course, students will be able to analyze the stability of fixed points and understand bifurcation concepts. They will develop skills in constructing and analyzing mathematical models for various real-world systems, including biological and ecological processes. Students will gain experience with discrete and continuous-time models, apply them to problems such as population dynamics and disease spread, and appreciate the role of mathematics in problem-solving and decision-making.				

Course Title		MATMJ610: Dynamical Systems			
Credits & Hours of Teaching		Theory	Practical	Cumulative	
	Credits	3	0	3	
	Hours of Teaching (Total)	45	0	45	
Course Objectives (not more than 100 words)	The course introduces the fundamental concepts of linear and non-linear dynamical systems. Topics include autonomous and non-autonomous systems, diagonalization, Jordan canonical forms, stability analysis, and the treatment of non-homogeneous systems. It also covers solutions to initial value problems, existence and uniqueness theorems, linearization, and the classification of critical points in phase space. The course explores applications of dynamical systems in various fields such as physics, biology, economics, and engineering.				
Units	Course Content				Teaching Hrs
I	Linear dynamical systems: preliminary concepts, autonomous and non-autonomous systems, diagonalization.				9
II	Fundamental theorem of linear systems.				9
III	Jordan canonical forms, stable, unstable, and center subspaces, nonhomogeneous linear systems.				9
IV	Non-linear dynamical systems: solutions to initial value problem, existence and uniqueness of solutions, linearization, phase space, classification of critical points.				9
V	Applications of Dynamical Systems. Applications in physics (e.g., pendulum, chaotic systems). Applications in biology (e.g., population dynamics). Applications in economics and engineering.				9
Texts / References	1. L. Perko: Differential equations and dynamical systems, Springer-Verlag, 2001. 2. F. Verhulst: Non-linear Differential Equations and Dynamical Systems, Springer, 1990.				
Learning	Students will gain a solid understanding of the theory and analysis of linear and non-linear				

Outcomes (not more than 100 words)	dynamical systems. They will learn to classify critical points, analyze the stability of systems, and apply mathematical techniques like diagonalization and linearization. Students will also develop the skills to solve initial value problems and explore practical applications in diverse areas, such as modeling population dynamics, analyzing chaotic systems, and solving engineering problems.				
Course Title	MATMJ611: Data Structures				
Credits & Hours of Teaching		Theory	Practical	Cumulative	
	Credits	2	1	3	
	Hours of Teaching (Total)	30	30	60	
Course Objectives (not more than 100 words)	This course comprehensively introduces fundamental data structures and algorithms in C programming. Students will learn about basic constructs such as arrays, strings, structures, and linked lists (including singly, circular, doubly, and circular doubly linked lists). The course covers stacks and queues, including their array and linked representations, operations, and evaluation of arithmetic expressions using Polish notations. Topics include hashing techniques, hash tables, hash functions, collision resolution methods, and their applications. Students will also explore trees (binary trees, binary search trees, AVL trees, red-black trees, multi-way search trees, heaps), and graphs, including graph terminology, traversal algorithms (BFS and DFS), topological sorting, minimum spanning trees, and shortest path algorithms.				
Units	Course Content				Teaching Hrs
I	C Fundamentals, Arrays, Strings, Structures, Linked Lists (Singly Linked Lists, Circular Linked Lists, Doubly Linked Lists, Circular Doubly Linked Lists).				6
II	Stacks, Array Representation of Stacks, Operations on a Stack, Linked Representation of Stacks, Operations on a Linked Stack, Evaluation of arithmetic expressions: Polish Notations, Queues, Array Representation of Queues, Linked Representation of Queues, Types of Queues (Circular Queues, Deques, Priority Queues).				6
III	Hashing and Collision: Motivation of Hashing, Hash Tables, Hash Functions, Different Hash Functions, Collision Resolution by Open Addressing, Collision Resolution by Chaining, Pros and Cons of Hashing, Applications of Hashing.				6
IV	Trees: Basic Terminology, Types of Trees, Creating a Binary Tree from a General Tree, Traversing a Binary Tree, Efficient Binary Trees (Binary Search Trees, AVL Trees, Red-black Trees), Multi-way Search Trees, Heaps. Graphs: Graph Terminology, Applications of Graphs.				6
V	Matrix Representation of Graphs, Graph Traversal Algorithms (Breadth-First Search (BFS) Algorithm and Depth-First Search (DFS) Algorithm), Topological Sorting, Minimum Spanning Trees (MST), MST Algorithms, Shortest Path Algorithms.				6
Texts / References	1. B.S. Gottfried: Schaum's Outline Programming with C, McGraw Hill, 1996. 2. T.H. Cormen: Charles E. Leiserson, Ronald L. Rivest, Clifford Stein, Introduction to Algorithms, 3rd Edition, The MIT Press, 2009. 3. A.M. Tenenbaum; Y. Langsam; M.J. Augenstein: Data Structures Using C, Pearson Education India, 2019. 4. D.B. West: Introduction to Graph Theory, 2nd Edition, Prentice Hall, 2000. 5. 5. Reema Thareja, Data Structures Using C, 2nd Edition, Oxford, 2014.				
Learning Outcomes (not more than 100 words)	By the end of this course, students will be proficient in implementing and using fundamental data structures such as arrays, strings, structures, linked lists, stacks, and queues. They will understand and apply various stack and queue operations, hashing techniques, and collision resolution methods. Students will also be skilled in working with different types of trees, including binary and balanced trees like AVL, red-black trees, multi-way search trees, and heaps. Additionally, they will master graph algorithms, including traversal (BFS and DFS), topological sorting, minimum spanning trees, and shortest path algorithms, and understand their applications.				

Course Title		MATMJ612: Analytical Visualization			
Credits & Hours of Teaching		Theory	Practical	Cumulative	
	Credits	1	2	3	
	Hours of Teaching (Total)	15	60	75	
Course Objectives (not more than 100 words)	This course introduces fundamental mathematical and geometric concepts, including elementary and special functions (such as the Heaviside and delta functions), asymptotes, curvature in 2D and 3D, and extrema. It covers Roll's Theorem, the Mean-Value Theorem, fixed-point theorems, fractals, and the Fibonacci sequence. Students will explore geometry topics like conic sections, spheres, cones, cylinders, and curves in $\mathbb{R}^3$ and study differential equations, including rotation of vectors, Kepler's laws, simple harmonic motion, and transforms (Laplace and Fourier). Mechanics topics include moment of inertia and angular momentum.				
Units	Course Content				Teaching Hrs
I	Elementary and Special functions (Step-side, Heaviside, delta, jump function), Asymptotes, Curvature (2D, 3D), Maxima-Minima, Tangent, and Normal, Roll's Theorem. Mean-Value theorem. Fixed Point Theorem, Fractals, Fibonacci sequence.				3
II	Convergence of sequence, Golden Ratio. Geometry: Conic Section, Sphere, Cone, Cylinder, Coincides, Right circular cylinder, Cardroid, Curve and surface in $\mathbb{R}^3$, Geodesics of Hyperbolic space, Sphere, Meshing. Differential Equation: Rotation of Vectors, Kepler's Law, S.H.M., ODE and its phase portrait, Bessel and Legendre functions, Laplace Transform, Fourier Transform, Fourier Series, PDE and Phase-Portrait. Mechanics: Moment of inertia, Compound Pendulum, Angular Momentum.				3
III	Basic course of simulation using MATLAB: MATLAB programming, Overview of MATLAB environment, commands and operations, Variables and data types, Scripts and functions, Basic plotting, Loops and execution control (if-else, switch-case, loops), Functions and their creation, Debugging and error handling, File input and output.				3
IV	Vectors and Matrices: Creating and manipulating vectors and matrices, Matrix operations (addition, subtraction, multiplication), Solving linear equations, Eigenvalues and Eigenvector, MATLAB function f-zero in a single variable, Fixed-point iteration in a single variable, Newton-Raphson in single variable and multiple variables, MATLAB function f-solve in single and multiple variables.				3
V	Advanced Plotting and Visualization: 2D and 3D plotting, customizing plots (labels, legends, titles, color map), Specialized plots (histograms, bar charts, scatter plots), Creating multiple plots in one figure (subplot, tiled layout), Plotting functions and data, Exporting and Printing Plots.				3
Texts / References	1. R. Pratap: Getting started with MATLAB: A Quick introduction for Scientists and Engineers, Oxford University Press Inc., 2002. 2. S. Attaway: A Practical Introduction to Programming and Problem Solving, Elsevier India Pvt Ltd-New Delhi, 2009. 3. B.H. Hahn; D.T. Valentini: Essential MATLAB For Engineers and Scientists Fourth Edition, Elsevier, 2010.				
Learning Outcomes (not more than 100 words)	By the end of this course, students will be proficient in mathematical and geometric concepts such as curvature, extrema, asymptotes, and differential equations. They will understand and apply fixed-point theorems, fractals, and sequences like Fibonacci. Students will be skilled in analyzing geometric shapes, differential equations, and transforms, including Laplace and Fourier. Additionally, they will gain hands-on experience with MATLAB, including programming basics, vector and matrix operations, solving equations, and advanced plotting techniques. They can effectively create, manipulate, and visualize mathematical data and functions using MATLAB tools.				

Course Title		MATMJ613: Graph Theory			
Credits &Hours of Teaching		Theory	Practical	Cumulative	
	Credits	3	0	3	
	Hours of Teaching (Total)	45	0	45	
Course Objectives (not more than 100 words)	This course aims to present a comprehensive understanding of advanced topics in graph theory. Students will learn to do calculations in graph and will understand the areas such as networks, the underlying algebra in graph theory and developing algorithms which are efficient. The course also is also intended to introduce students to various topics which are often omitted from a first course on graph theory. Students will also be exposed to real world problems which are being/have been solved by the help of graph theory.				
Units	Course Content				Teaching Hrs
I	The Introduction: Graphs, Degree of a vertex, Paths and Cycles, Connectivity, Examples of Graphs such as bipartite graphs, Planar and non planar graphs, Contraction, Minor, Trees and Forests, distances				9
II	Matching Covering and Packing: Matching, in a graph and matching in a bipartite graph. Assignment problems, Covering and Packing, Covering by paths and cycles, Tutte’s theorem				9
III	Coloring: Edge and vertex colorings, 4 and 5 colorable graphs, List coloring, Perfect graphs. Flows: Flow in a network, group valued flows, Tutte’s flow conjecture, k-Flows, Supply and demand problems				9
IV	Extremal graph theory: Minors, Hadwigers Conjecture, Regularity, Szemerédi’s Regularity Lemma, Using the regularity lemma, partitions using paths and cycles. Ramsey Theory: Ramser Numbers, Induced Ramsey Theorem, Ramsey property and connectivity, Sperner’s Lemma				9
V	Random Graphs: Definition, Existence and expectation, probabilistic methods, properties of almost all graphs, Threshold function.				9
Texts / References	1. R Diestel: Graph Theory GTM 173 IV ed., Springer 2012. 2. J A Bondy, USR Murty: Graph Theory, GTM 244, Springer, 2008. 3. Douglas West: Introduction to graph theory, Pearson, 2000.				
Learning Outcomes (not more than 100 words)	By the end of the course, students will have a good understanding of complexity and algorithms. They will also be able to relate and model the relational real world problem in terms graphs. The subject graph theory, like number theory has its own flavor and the students will be able to appreciate and understand this on completion of this course. This course will be very useful for both the areas which are referred to pure and applied Mathematics. On completion of this course students may be easily inducted into research in graph theory.				

Course Title		MATMJ614: Vedic Mathematics			
Credits & Hours of Teaching		Theory	Practical	Cumulative	
	Credits	3	0	3	
	Hours of Teaching (Total)	45	0	45	
Course Objectives (not more than 100 words)	This course offers an in-depth exploration of Vedic Mathematics, highlighting its historical significance and foundational principles. Students will engage with the system's distinctive sutras and sub-sutras, learning to apply them to various arithmetic operations, algebraic, geometric, and contributions of Indian Mathematician. The curriculum emphasizes rapid calculation approach, covering multiplication, squaring, cubing, square root, cube root, multiplication and division factorization and solution of equations.				
Units	Course Content				Teaching Hrs
I	Multiplication in arithmetic: <i>Nikhilam</i> sutra, <i>Antyayordasake'pi</i> sutra Multiplication: <i>Urdhvatiryak</i> Concept of <i>bijanka</i> , and Verification of all the operations by <i>Bijanka</i> , Square in arithmetic by Yavadunam, <i>Ekadhikena</i> sutra, <i>Dwandvayoga</i> sutra, Cube in arithmetic by sutras <i>Nikhilam</i> and <i>Anurupyena</i> sutra, Square root of four or -five-digit numbers Cube root Applications of Meru-prastar in higher order powers.				9
II	Decimal and fraction, division by <i>Nikhilam</i> sutram, division of $1/19$, $1/29$ by <i>Ekadhikena</i> sutram, division by <i>Paravartya</i> sutram, division by <i>Anurupyena</i> sutram, division of polynomials, factors of general second-degree equation by <i>Lopsthapanabhyam</i> sutram.				9
III	Multiplication of linear, quadratic, and cubic polynomials in one variable. (<i>Urdhvatiryak</i> sutra), Division of polynomials by <i>Sutra-Paravartya Yojayet</i> , Solution of particular types of linear equations in one variable, Factors of the quadratic and Cubic Expressions, Solutions of simultaneous linear in two variables.				9
IV	A brief introduction to the four Shulva sutras, Baudhayan triplets for the angle of a right-angled triangle, Addition and subtraction of Baudhayan triplets, Concept, definition, and properties of six trigonometric functions based on Baudhayan triplets. Formulation of Identities, Heights and distance by Baudhayan.				9
V	Introduction of Sutras and sub-sutras by Swami Bharati Krishna Tirtha ji, Contributions of Aryabhata, Life and contributions of Bhaskara II. Kaparekar Numbers, Life and contributions of Ramanujan.				9
Texts / References	1. S.B. Krishna Tirtha, Vedic Mathematics, Motilal Banarasi Das, 1965. 2. Vedic Ganita: Vihangam Drishti-I, Shiksha Sanskriti Utthan Nyasa, New Delhi. 3. V.B. Datta; A.N. Singh: History of Hindu Mathematics A Source Book Parts I and II, 1935. 4. V.G. Unkalkar: Magical World of Mathematics (Vedic Mathematics) Vandana Publishers Bangalore, 2017. 5. V.G. Unkalkar: Excel With Vedic Mathematics, Vandana Publishers Bangalore, 2019.				
Learning Outcomes (not more than 100 words)	By the end of the course, students will be proficient in applying Vedic Mathematics for high-speed calculations, such as addition, subtraction, and multiplication. They will master techniques for squaring and cubing numbers, working with decimals and fractions, and solving equations. Students will gain skills in mental arithmetic and a deeper understanding of Vedic mathematical concepts.				

Course Title		MATMV61: Formal Languages and Automata Theory			
Credits & Hours of Teaching		Theory	Practical	Cumulative	
	Credits	3	1	4	
	Hours of Teaching (Total)	45	30	75	
Course Objectives (not more than 100 words)	This course aims to provide a deep understanding of the theoretical foundations of computer science, focusing on formal languages, automata, and computability. Students will explore the Chomsky hierarchy, learn to design and analyze various types of automata and understand the computational limits defined by Turing machines. The course also covers complexity theory, emphasizing the practical applications of formal languages and automata in software engineering, compiler design, and network protocols.				
Units	Course Content				Teaching Hrs
I	Introduction: Alphabet, languages and grammars, productions and derivation, Language generated by a grammar, Chomsky hierarchy. Regular languages and finite automata: Deterministic and non-deterministic finite automata, acceptance of language by finite automata, properties of regular languages, equivalence and minimization of finite automata, Moore and Mealy machines, regular expressions, regular grammars, Pumping lemma.				9
II	Context-free and context-sensitive languages: Context-free grammars (CFGs) and languages (CFLs), parse trees, ambiguity in CFG, Chomsky and Greibach normal forms, Pumping lemma for CFLs, closure properties of CFLs, Pushdown automata; Context-sensitive grammars and languages, linear bounded automata. Recursively enumerable				9
III	languages and Turing machines: The basic model for Turing machines, configurations and acceptance, Turing-recognizable (recursively enumerable) and Turing-decidable (recursive) languages, unrestricted grammars, universal Turing machines, halting problem.				9
IV	Regular Languages, Context-Free Languages, Context-Sensitive Languages, Recursively Enumerable Languages, Linear Bounded Automata, Examples and Applications of Chomsky Hierarchy, Advanced Topics in Automata Theory, Minimization of Finite Automata, Myhill-Nerode Theorem, Introduction to Quantum Automata, Automata in Real-World Applications: Compilers, Parsing, Network Protocols.				9
V	Computability and Complexity, Introduction to Complexity Classes: P, NP, and NP-Complete, Reductions and Cook-Levin Theorem, Time Complexity and Space Complexity, The Role of Automata in Complexity Theory, Applications and Case Studies, Applications of Automata and Formal Languages in Software Engineering, Case Studies: Lexical Analysis, Compiler Design, Model Checking, Current Research Trends in Formal Languages and Automata Theory				9
Texts / References	1. D.C. Kozen: Automata and Computability, Undergraduate Texts in Computer Science, Springer, 1997. 2. J.E. Hopcroft; R. Motwani; J.D. Ullman, Introduction to Automata Theory, Languages, and Computation, third edition, Prentice Hall, 2007. 3. H.R. Lewis; C.H. Papadimitriou: Elements of the Theory of Computation, second edition, Prentice Hall, 1998. 4. M. Sipser: Introduction to the Theory of Computation, third edition, Course Technology, 2005. 5. P. Linz: An Introduction to Formal Languages and Automata, Narosa, 2007. 6. D.I.A. Cohen: Introduction to Computer Theory, second edition, Wiley, 1991.				
Learning Outcomes (not more than 100 words)	Upon completing this course, students will be able to analyze and classify languages using the Chomsky hierarchy; Design and minimize finite automata, pushdown automata, and Turing machines; Apply formal language concepts to solve real-world problems in software engineering and computational theory; Understand and explain key concepts in computability, complexity theory, and their implications for computer science; Explore advanced topics in automata theory and current research trends.				

Course Title		MATMJ71/MATMN71: Topology			
Credits &Hours of Teaching		Theory	Practical	Cumulative	
	Credits	4	0	4	
	Hours of Teaching (Total)	60	0	60	
Course Objectives (not more than 100 words)	This course introduces fundamental concepts in topology, covering basic notions such as open and closed sets, closure, interior, and limit points. Students will explore continuous maps, homeomorphisms, and different types of maps, including open and closed maps. The course will examine quotient topology and identification spaces, including the Möbius strip, projective plane, torus, and Klein bottle. Key topics include connectedness, locally connected spaces, components, path connectedness, compactness, and separation axioms (T_0 , T_1 , T_2 , T_3). Students will study Urysohn's lemma, Tietze's extension theorem, Urysohn's embedding, metrization theorem, countability axioms, compactifications, and paracompactness. Additionally, the course covers the homotopy of maps and paths, the fundamental group, covering spaces and transformations, simply connectedness, the universal cover, and Van Kampen's theorem.				
Units	Course Content				Teaching Hrs
I	Open sets. Neighborhoods. Interior,Exterior, and Boundary. Limit points and Derived set. Closed sets and Closure. Bases and Subbases.				12
II	Continuous Mappings. Homeomorphism. Induced and coinduced topologies. Subspace topology, Product topology, and Quotient topology. Identification Spaces (Cone, Möbius Strip, Projective plane, Torus, Klein bottle etc.).				12
III	Separable spaces, Lindelöf spaces, First countable spaces and Second countable spaces. Cantor-Bendixson Theorem. Compactness, Compactifications and Paracompactness.				12
IV	Separation axioms: Hausdorff space, Regular spaces, Completely regular spaces,Tychonoff spaces, Normal spaces, T_4 space.Urysohn's lemma, Tietze's extension theorem, Urysohn's embedding and Metrization theorem.Connectedness, Locally Connected Spaces, Components, Path connectedness andLocally path connectedness.				12
V	Homotopy of maps and paths, Fundamental group, Covering Spaces, Covering transformations, Simply Connectedness, Universal Cover, Van Kampen Theorem.				12
Texts / References	1. M.A. Armstrong:Basic Topology, Springer Verlag, 1983. 2. M.H. Cho: Discovering modern general topology, Kyung Moon Publishers, Korea, 2001. 3. F.H. Croom: Principles of Topology, Kyung Moon Publishers, Korea 1996. 4. J. Dugundji: Topology, Prentice-Hall. 1966. 5. J.L. Kelly:General Topology, van Nostrand, Reinhold Co., New York, 1995. 6. J.R. Munkres: Topology Prentice Hall of India, 2000. 7. B.L. Sharma: Sanstithi I - Sāmānya Sanstithi (Topology I - General Topology), Uttar Pradesh Hindi Sansthan, Hindi Granth Academy Division, Lucknow, 1980. 8. R. Vaidyanathaswamy:Treatise on set topology, Indian mathematical society, 1947.				
Learning Outcomes (not more than 100 words)	By the end of this course, students will be able to understand and apply fundamental topological concepts, including open and closed sets, continuity, and homeomorphisms. They will grasp quotient topology and identification spaces, such as the Möbius strip and Klein bottle. Students will analyze connectedness, compactness, and separation axioms and apply Urysohn's lemma, Tietze's extension theorem, and metrization theorems. They will also explore countability axioms, compactifications, and paracompactness. In algebraic topology, students will understand homotopy, the fundamental group, covering spaces, and covering transformations, and will apply Van Kampen's theorem to compute fundamental groups.				

Course Title	MATMJ72: Advanced Algebra				
Credits & Hours of Teaching		Theory	Practical	Cumulative	
	Credits	4	0	4	
	Hours of Teaching (Total)	60	0	60	
Course Objectives (not more than 100 words)	This course aims to provide an in-depth exploration of group theory and ring theory. Students will learn about isomorphism theorems, symmetric and alternating groups, dihedral groups, matrix groups, and isometry groups of $\mathbb{R}^2$ and $\mathbb{R}^3$. The course covers direct products, indecomposable groups, group actions, stabilizer subgroups, orbit decomposition, and the class equation. Key concepts include Sylow subgroups, Sylow's theorems, p-groups, and their applications. In ring theory, topics include rings, subrings, homomorphisms, ideals, quotient rings, integral domains, polynomial rings, field of fractions, division rings, fields, and subfields. Students will explore field extensions, factorization of polynomials, splitting fields, separable extensions, automorphisms, normal extensions, Galois theory, finite fields, cyclic Galois groups, the fundamental theorem of algebra, cyclotomic extensions, and solvability by radicals.				
Units	Course Content				Teaching Hrs
I	Isomorphism theorems for groups, Symmetric groups, Alternating groups, Dihedral groups, Matrix groups, Isometry groups of $\mathbb{R}^2$ and $\mathbb{R}^3$, Internal and External direct product and their relationship, Indecomposable groups.				12
II	Action of a group G on a set, Stabilizer subgroups and orbit decomposition, Class equation of an action, Sylow subgroups, Sylow's Theorem I, II and III, p-groups, Examples and applications, Groups of order p q, Direct and inverse images of Sylow subgroups.				12
III	Rings, Examples and properties, Subrings, Homomorphism, Ideals and Quotient rings, Integral domain, Polynomial rings, Field of fraction, Division rings, Fields, subfields. Characteristic of a ring and field, Prime subfields, Field extensions, Finite extensions, Algebraic and transcendental extensions.				12
IV	Factorization of polynomials in extension fields. Splitting fields. Separable extensions, Perfect fields. Automorphisms of fields, Normal extensions, normal closures, Galois extensions, Fundamental theorem of Galois theory, Computation of Galois groups of polynomials.				12
V	Primitive element, Finite fields, Subfields of finite fields, Cyclic Galois groups, fundamental theorem of algebra. Cyclotomic extensions and polynomials, Cyclic extensions, Solvability by radicals.				12
Texts / References	1. D.S Dummit; R. Foote: Abstract Algebra, Wiley, 2003. 2. J. Gallian: Contemporary Abstract Algebra, Narosa, 1998. 3. R. Lal: Algebra vol 1 and 2, Springer, 2022. 4. J.B. Fraleigh: A first course in Abstract Algebra, Pearson, 2002.				
Learning Outcomes (not more than 100 words)	By the end of this course, students will be proficient in group theory, including isomorphism theorems, symmetric, alternating, and dihedral groups, matrix groups, and group actions. They will understand stabilizer subgroups, orbit decomposition, Sylow subgroups, and Sylow's theorems. In ring theory, students will be adept at working with rings, subrings, ideals, quotient rings, polynomial rings, and fields. They will grasp concepts of field extensions, factorization of polynomials, separable and normal extensions, automorphisms, and Galois theory. Additionally, students will analyze finite fields, cyclic Galois groups, cyclotomic extensions, and the solvability of polynomials by radicals.				

Course Title		MATMJ73: Functional Analysis			
Credits & Hours of Teaching		Theory	Practical	Cumulative	
	Credits	4	0	4	
	Hours of Teaching (Total)	60	0	60	
Course Objectives (not more than 100 words)	This course aims to provide a thorough understanding of normed linear and Banach spaces, completeness of $\square_\square$ space. Students will explore bounded linear transformations, normed linear spaces, and fundamental theorems like the open mapping, closed graph, and uniform boundedness principle. The course will also address the Hahn-Banach theorem, dual spaces, separability, reflexive spaces, weak convergence, and compact operators. Additionally, it will delve into inner product spaces, Hilbert spaces, orthonormal sets, and essential theorems and operators in Hilbert spaces, including the Riesz representation theorem and self-adjoint operators.				
Units	Course Content				Teaching Hrs
I	Normed linear spaces and Banach spaces. The $\square_\square$ -space as Banach spaces				12
II	Bounded linear transformations. $\square(\square, \square)$ as a normed linear space. Open mapping and closed graph theorems. Uniform boundedness principle and its consequences. Hahn-Banach theorem and its application				12
III	Dual spaces with examples. Separability. Reflexive spaces. Weak convergence. Compact operators.				12
IV	Inner product spaces, Hilbert spaces. Orthonormal sets. Bessel's inequality. Complete orthonormal sets and Parseval's identity. Structure of Hilbert spaces. Projection theorem.				12
V	Riesz representation theorem. Riesz-Fischer theorem. Adjoint of an operator on a Hilbert space. Reflexivity of Hilbert spaces. Self-adjoint operators. Positive, projection, normal, and unitary operators.				12
Texts / References	1. H.L. Royden: Real Analysis, Macmillan, 4th Edition, 1993. 2. G.F. Simmons: Introduction to Topology and Modern Analysis, McGraw-Hill, 1963. 3. G. Bachman; L. Narici: Functional Analysis, Academic Press, 1966. 4. A.E. Taylor: Introduction to Functional Analysis, John Wiley, 1958. 5. B.V. Limaye: Functional Analysis, Wiley Eastern. 6. N. Dunford; J.T. Schwartz: Linear Operators, Part-I, Interscience, 1958. 7. R.E. Edwards: Functional Analysis, Holt Rinehart and Winston, 1965. 8. C. Goffman; G. Pedrick: First Course in Functional Analysis, Prentice- Hall of India, 1987. 9. K.K. Jha: Functional Analysis and Its Applications, Students' Friend, 1986. 10. Erwin Kreyszig, Introductory Functional Analysis with Applications, John Wiley and Sons, 1978. 11. Rajendra Bhatia, Notes on Functional Analysis, Hindustan Book Agency (Springer), 2009.				
Learning Outcomes (not more than 100 words)	By the end of this course, students will understand and work with normed linear spaces, Banach spaces, and $\square_\square$ -spaces, applying key inequalities and concepts of convergence. They will handle bounded linear transformations and normed linear spaces, utilizing the open mapping, closed graph, and uniform boundedness theorems. Students will apply the Hahn-Banach theorem, understand dual and reflexive spaces, and analyze weak convergence and compact operators. They will gain proficiency in inner product spaces, Hilbert spaces, orthonormal sets, and essential theorems such as Riesz representation and Riesz-Fischer. Additionally, students will work with various operators in Hilbert spaces, including self-adjoint operators.				

Course Title		MATMJ74: Differentiable Manifolds			
Credits &Hours of Teaching		Theory	Practical	Cumulative	
	Credits	4	0	4	
	Hours of Teaching (Total)	60	0	60	
Course Objectives (not more than 100 words)	This course aims to provide a rigorous introduction to the concepts of differentiable manifolds and related topics in differential geometry. Students will explore the theory of differentiable maps, tangent spaces, vector fields, and integral curves. The course covers advanced topics like submanifolds, distributions, integrability, tensor calculus, and connections, including the Koszul connection. Additionally, the course delves into the geometry of Lie groups, Lie algebras, and fibre bundles, offering insights into their applications in physics and mathematics.				
Units	Course Content				Teaching Hrs
I	Differentiable manifolds - Definitions and examples. Differentiable maps. Jacobian matrix. Tangent spaces. Vector fields. Integral curves.				15
II	Submanifolds. Distributions and integrability. 1-parameter group of transformations. Tensors and forms.				15
III	The Koszul connection. Covariant, Lie and Exterior derivatives. Geodesics and curvature.				15
IV	Lie Groups and Lie Algebra. Fibre bundles				15
Texts / References	1. R.L.Bishop; R.J.Crittenden:GeometryofManifolds,AcademicPress,1964. 2. S.S.Chern;W.H.Chen; K.S.Lam:LecturesonDifferentialGeometry,WorldScientific,2000. 3. N.J.Hicks:NotesonDifferentialGeometry,VonNostrand,1965. 4. J.M.Lee:Introduction toSmoothManifolds,Springer,2006. 5. Y.Matsushima: DifferentiableManifolds, Dekker,1972. 6. M.Spivak: AComprehensiveIntroductiontoDifferential GeometryVol 13ed., PublisherPerish,1999. 7. F.Warner:FoundationsofDifferentiableManifoldsandLieGroups,Springer,1983.				
Learning Outcomes (not more than 100 words)	Upon completion of this course, students will be able to:understand and apply the fundamental concepts of differentiable manifolds and tangent spaces; analyze and compute the Jacobian matrix, vector fields, and integral curves; apply the theory of connections, including the Koszul connection, to study geodesics and curvature; grasp the structure and properties of Lie groups, Lie algebras, and fibre bundles; use tensor calculus and differential forms to solve complex problems in differential geometry.				

Course Title		MATMJ81/MATMJ8R1: Geometric Topology			
Credits & Hours of Teaching		Theory	Practical	Cumulative	
	Credits	4	0	4	
	Hours of Teaching (Total)	60	0	60	
Course Objectives (not more than 100 words)	This course introduces advanced topics in topology, focusing on simplicial complexes, including n-simplices, simplicial maps, and triangulations. Students will explore affinely and geometrically independent points, and the geometric carrier and realizations of simplicial complexes. The course covers singular simplices, chains, cycles, and boundaries, leading to understanding singular homology and its axioms. Students will compute the homology of various surfaces like spheres, projective planes, tori, Möbius strips, and Klein bottles. The course also addresses the classification of closed surfaces, Euler characteristics, and surface groups. Additionally, students will examine the classification problem in 3-manifolds, including Poincaré's theorem, loop and sphere theorems, prime decompositions, and the study of Seifert fibered spaces and hyperbolic manifolds.				
Units	Course Content				Teaching Hrs
I	Affinely and geometrically independent points, n-Simplex, Simplicial complex in dimension n, simplicial maps, geometric carrier and realizations, link and star of a simplex, Triangulations. Examples and computations of Triangulations representing Topological objects.				12
II	Singular simplices, chains, cycles and boundaries. Group of chains and boundaries, Singular Homology. Discussion of various axioms. Computation of Homology of surfaces such as Sphere, Projective plane, Torus, Mobius strip, Cylinder, Klein bottle.				12
III	Orientable double cover. Euler Characteristics, closed surfaces as quotients of 4g-gon. Discussion of classifications of closed surfaces. Surface groups and presentation. Two dimensional geometries.				12
IV	Discussion of Classification problem in Topology with an Overview of Poincaré Theorem and Geometrization in dimension 3. Non availability of strong invariants in dimension 3. Various examples of 3-manifolds. Loop theorem, sphere theorem.				12
V	Prime decomposition, irreducibility. Torus decompositions. JSJ, Seifert fibered spaces and Hyperbolic Manifolds. Non solvability of word problem: Manifolds of dimension 4 and more.				12
Texts / References	1. A. Hatcher: Algebraic Topology, Cambridge Univ Press, 2002. 2. A. Hatcher: Basic, Topology of 3-manifolds. 3. J. Hempel: 3-Manifolds. Annals of Math Studies 86. Princeton University Press, 1976. 4. V.V. Prasolov; A.B. Sossinsky: Knots, Links, Braids and 3-Manifolds, AMS, 1997. 5. J.R. Munkres: Elements of Algebraic Topology, CRC Press, 1984.				
Learning Outcomes (not more than 100 words)	By the end of this course, students will be proficient in working with simplicial complexes, including computing triangulations and understanding simplicial maps and geometric realizations. They will master concepts such as singular simplices, chains, cycles, and boundaries, and will be able to compute and analyze singular homology for surfaces like spheres and tori. Students will also understand the classification of closed surfaces, Euler characteristics, and surface groups. Additionally, they will be familiar with key results in 3-manifold topology, including the Poincaré theorem, prime decompositions, and the study of hyperbolic and Seifert fibered spaces.				

Course Title		MATMJ82/MATMJ8R2: Fourier Analysis			
Credits & Hours of Teaching		Theory	Practical	Cumulative	
	Credits	4	0	4	
	Hours of Teaching (Total)	60	0	60	
Course Objectives (not more than 100 words)	This course explores fundamental aspects of Fourier series and transforms. Students will study the uniqueness of Fourier series, convolutions, and summability methods such as Cesaro and Abel. Key topics include Fejer's theorem, the Poisson kernel, and the Dirichlet problem in the unit disc. The course will address mean square convergence and provide continuous functions with divergent Fourier series examples. In distributions, students will learn about the calculus of distributions, Schwartz class functions, and the Fourier transforms of rapidly decreasing functions. They will also cover the Riemann-Lebesgue lemma, the Fourier inversion theorem, and the Fourier transforms of Gaussians and tempered distributions.				
Units	Course Content				Teaching Hrs
I	Basic Properties of Fourier Series: Uniqueness of Fourier Series, Convolutions, Cesaro and Abel Summability, Fejer's theorem.				12
II	Poisson Kernel and Dirichlet problem in the unit disc. Mean square Convergence, Example of Continuous functions with divergent Fourier series.				12
III	Distributions and Fourier Transforms: Calculus of Distributions, Schwartz class of rapidly decreasing functions.				12
IV	Fourier transforms of rapidly decreasing functions, Riemann Lebesgue lemma, Fourier Inversion Theorem, Fourier transforms of Gaussians.				12
V	Tempered Distributions: Fourier transforms of tempered distributions, Convolutions.				12
Texts / References	1. R.S. Strichartz: A Guide to Distributions and Fourier Transforms, CRC Press, 1994. 2. E.M. Stein; R. Shakarchi: Fourier Analysis: An Introduction, Princeton University Press, 2003. 3. I. Richards; H. Youn: Theory of Distributions: A Nontechnical Introduction, Cambridge University Press, 1990. 4. R.S. Pathak: A Course in Distribution Theory and Applications, Narosa, 2001. 5. G. H. Hardy, Divergent Series, Oxford University Press, 1973. 6. A. Zygmund, Trigonometric Series, Cambridge University Press, 2002.				
Learning Outcomes (not more than 100 words)	By the end of this course, students will be proficient in analyzing and applying the Fourier series, including understanding uniqueness, convolutions, and various summability methods. They will be able to use Fejer's theorem and the Poisson kernel and solve the Dirichlet problem in the unit disc. Students will also grasp mean square convergence and handle examples of continuous functions with divergent Fourier series. In distribution theory, they will understand the calculus of distributions, the Schwartz class, and Fourier transforms of rapidly decreasing functions. They will apply the Riemann-Lebesgue lemma and the Fourier inversion theorem and handle Fourier transforms of Gaussians and tempered distributions.				

Course Title		MATMJ83/MATMJ8R3: Nonlinear Programming			
Credits & Hours of Teaching		Theory	Practical	Cumulative	
	Credits	4	0	4	
	Hours of Teaching (Total)	60	0	60	
Course Objectives (not more than 100 words)	This course provides a comprehensive introduction to convex analysis and optimization techniques. Students will study convex sets, convex combinations, Carathéodory's theorem, convex hulls, separation theorems, and convex functions, including their epigraphs and lower level sets. Key topics include the continuity of convex functions, differentiable and twice differentiable convex functions, and generalized convexity. The course covers unconstrained optimization methods, such as first and second order necessary conditions, line search techniques, and methods including the golden section, Fibonacci, quadratic and cubic interpolation, steepest descent, Newton's method, conjugate gradient, and quasi-Newton methods. In constrained optimization, students will explore local, global, and isolated minimizers, feasible and descent directions, Lagrangian functions, Fritz-John and Karush-Kuhn-Tucker conditions, convex programming, duality, and saddle points.				
Units	Course Content				Teaching Hrs
I	Convex analysis: convex set, convex combination, Caratheodory's theorem, convex hull, Separation theorems, convex functions, epigraphs, lower level sets, continuity of convex functions.				12
II	Alternative theorems, differentiable and twice differentiable convex functions, generalized convexity.				12
III	Unconstrained Optimization: local and global minimizers, first and second order necessary optimality conditions, second order sufficient optimality condition, unimodal functions, line search, golden section method.				12
IV	Fibonacci Method, Quadratic Interpolation Methods, Cubic Interpolation Method, the steepest descent method, the Newton method, the conjugate gradient method, Quasi-Newton method.				12
V	Constrained Optimization: local, global and isolated minimizers, feasible directions, descent directions, Lagrangian function, Fritz-John optimality conditions, constraint qualifications, Karush-Kuhn-Tucker optimality conditions, convex programming, second order optimality conditions, duality, saddle points.				12
Texts / References	1. O.L. Mangasarian: Nonlinear programming. Society for Industrial and Applied Mathematics, 1994. 2. J. Nocedal; S.J. Wright: Numerical optimization. New York, NY: Springer New York, 1999. 3. S.P. Boyd; L. Vandenberghe: Convex optimization. Cambridge University Press, 2004. 4. S.K. Mishra; B. Ram: Introduction to unconstrained optimization with R. Springer Nature, 2019 5. E.K. Chong; W.S. Lu; S.H. Zak: An Introduction to Optimization: With Applications to Machine Learning. John Wiley and Sons, 2023				
Learning Outcomes (not more than 100 words)	By the end of this course, students will be able to analyze and apply concepts from convex analysis, including convex sets, functions, and separation theorems. They will understand the continuity of convex functions and generalized convexity. In unconstrained optimization, students will solve problems using various methods such as line search, quadratic and cubic interpolation, and gradient-based techniques including steepest descent, Newton's method, and conjugate gradient methods. For constrained optimization, students will apply Lagrangian methods, Fritz-John and Karush-Kuhn-Tucker conditions, and handle duality and saddle points, addressing both convex and general programming problems.				

Course Title		MATMJ84/MATMJ8R4: Quadratic Programming			
Credits & Hours of Teaching		Theory	Practical	Cumulative	
	Credits	4	0	4	
	Hours of Teaching (Total)	60	0	60	
Course Objectives (not more than 100 words)	This course focuses on quadratic programming problems, including unconstrained and constrained cases with equality and inequality constraints. Students will study quadratic cost functions, convex quadratic functions, and concepts related to local and global minimizers. The course covers existence and uniqueness theorems, KKT systems, and min-max dual and saddle point problems. Students will analyze the properties of solution sets, solution map continuity, and apply Farka's Lemma. The course also addresses duality in constrained quadratic programming, numerical algorithms, nonconvex quadratic programming, and applications in portfolio optimization and affine variational inequalities.				
Units	Course Content				Teaching Hrs
I	Unconstrained quadratic programming problems, Quadratic cost functions, Convex quadratic functions, Local and Global minimizers, Existence and Uniqueness theorems.				12
II	Quadratic Programming problems with equality constraints, Optimality, Existence and Uniqueness, KKT systems, Min-max, Dual and saddle point problems, Properties of the solution sets of quadratic.				12
III	Programming problems, Continuity of the solution map, Quadratic programming problems with inequality constraints.				12
IV	Farka's Lemma, Optimality, Existence and Uniqueness of solution, Duality for constrained quadratic programming problems, Algorithms for numerical solutions of quadratic programming problems.				12
V	Nonconvex quadratic programming problems, Applications to Portfolio optimization and Affine Variational inequalities.				12
Texts / References	1. Z. Dostal: Optimal Quadratic Programming Algorithms, Springer, New York, 2009. 2. M.J. Best: Quadratic Programming with Computer Programs, CRC, 2017. 3. G.M. Lee; N.N. Tam; N.D. Yen: Quadratic Programming and Affine Variational Inequalities, Springer, New York, 2005.				
Learning Outcomes (not more than 100 words)	By the end of this course, students will be able to solve unconstrained and constrained quadratic programming problems, including those with equality and inequality constraints. They will understand quadratic cost functions, convexity, and optimality conditions and be able to apply KKT systems, min-max duality, and saddle point concepts. Students will analyze solutions' existence, uniqueness, and continuity, and utilize Farka's Lemma. They will also be proficient in duality theory and numerical algorithms for solving quadratic programming problems, and they will apply these techniques to real-world scenarios such as portfolio optimization and affine variational inequalities.				

Course Title	MATMJ85/MATMJ8R5: Fluid Dynamics				
Credits & Hours of Teaching		Theory	Practical	Cumulative	
	Credits	4	0	4	
	Hours of Teaching (Total)	60	0	60	
Course Objectives (not more than 100 words)	This course aims to comprehensively understand fluid dynamics and its mathematical foundations. Key objectives include exploring fundamental concepts such as the equation of continuity, boundary surfaces, and the nature of fluid flow through various geometries. Students will study irrotational and rotational motions, vortex lines, and key theorems like Euler's Equation of Motion and Bernoulli's Theorem. Additionally, the course covers stress components in real fluids, principal stresses, and the transformation of stress components. Practical applications include analyzing flows between flat plates, through pipes, and between rotating cylinders.				
Units	Course Content				Teaching Hrs
I	Equation of continuity, Boundary surfaces, path lines and streamlines, Irrotational and rotational motions, Vortex lines, Euler's Equation of motion, Bernoulli's theorem, Impulsive actions, Motion in two-dimensions.				12
II	Conjugate functions, Source, sink, doublets and their images, conformal mapping. Stress components in real fluid, Equilibrium equation in terms of stress components, Transformation of stress components.				12
III	Principal stresses, Nature of strains, Transformation of rates of strain, Relationship between stress and rate of strain.				12
IV	Navier-Stokes equation of motion. Buckingham II-theorem, Flow between parallel flat plates, Couette and plane Poiseuille flows.				12
V	Flow through a pipe, Hagen Poiseuille flow, flow between two co-axially cylinders and two concentric rotating cylinders, Unsteady motion of a flat plate.				12
Texts / References	1. W.H. Besant; A.S. Ramsey: A Treatise on Hydrodynamics, CBS Publishers and Distributors, Delhi, 1988. 2. S.W. Yuan: Foundations of Fluid Dynamics, Prentice-Hall of India, 1988. 3. F. Charlton: A Text Book of Fluid Dynamics, CBC, 1985.				
Learning Outcomes (not more than 100 words)	Upon completion of the course, students will be able to apply fluid dynamics principles to analyze and solve problems involving fluid flow, including calculating stress components and understanding the behavior of fluids under various conditions. They will master conjugate functions, source and sink dynamics, and conformal mapping. Students will be proficient in applying the Navier-Stokes equation, Buckingham II-theorem, and analyzing flows in different geometries like pipes and rotating cylinders. They can also perform calculations related to unsteady motion and boundary conditions.				

Course Title		MATMJ86/MATMJ8R6: Numerical Analysis			
Credits & Hours of Teaching		Theory	Practical	Cumulative	
	Credits	4	0	4	
	Hours of Teaching (Total)	60	0	60	
Course Objectives (not more than 100 words)	This course aims to equip students with a comprehensive understanding of integral equations and matrix computations, and their application in numerical methods for differential equations. Key objectives include mastering Fredholm and Volterra integral equations, converting initial and boundary value problems into integral equations, and solving them using numerical methods like Newton-Cotes, Lagrange interpolation, and Chebyshev polynomials. Students will also learn techniques for solving systems of linear equations, matrix inversion, and factorization. The course covers numerical methods for first-order and second-order differential equations, boundary value problems, and introduces the Finite Element Method for solving boundary value problems.				
Units	Course Content				Teaching Hrs
I	Integral equations: Fredholm and Volterra equations of first and second types. Conversions of initial and boundary value problems into integral equations, numerical solutions of integral equations using Newton-Cotes, Lagrange's linear interpolation and Chebyshev polynomial.				12
II	Matrix Computations: System of linear equations, Conditioning of Matrices, Matrix inversion method, Matrix factorization, Tridiagonal systems.				12
III	Numerical solutions of system of simultaneous first order differential equations and second order initial value problems (IVP) by Euler and Runge-Kutta (IV order) explicit methods.				12
IV	Numerical solutions of second order boundary value problems (BVP) of first, second and third types by shooting method and finite difference methods.				12
V	Finite Element method: Introduction, Methods of approximation: Rayleigh-Ritz Method, Galerkin Method and its application for solution of BVP.				12
Texts / References	1. M.K. Jain; S.R.K. Iyenger; R.K. Jain: Numerical Methods for Scientific and Engineering Computations, New Age Publications, 2003. 2. M.K. Jain: Numerical Solution of Differential Equations, 2nd edition, Wiley-Eastern. 3. D.V. Griffiths; I.M. Smith: Numerical Methods for Engineers, Oxford University Press, 1993. 4. S.S. Sastry: Introductory Methods of Numerical Analysis. 5. C.F. Gerald; P.O. Wheatley: Applied Numerical Analysis, Addison- Wesley, 1998. 6. S. Gupta: Text Book on Calculus of Variation, Prentice-Hall of India, 2002. 7. N. Kumar: An Elementary Course on Variational Problems in Calculus, Narosa Publishing House, New Delhi, 2004.				
Learning Outcomes (not more than 100 words)	Upon completion, students will proficiently solve Fredholm and Volterra integral equations and convert boundary and initial value problems into integral equations. They will effectively use numerical methods, such as Euler and Runge-Kutta methods, to solve differential equations. Students will gain expertise in matrix computations, including matrix inversion and factorization. They will also be able to solve boundary value problems using shooting and finite difference methods and apply the Finite Element Method, including Rayleigh-Ritz and Galerkin methods, to solve complex boundary value problems.				

Course Title		MATMJ87/MATMJ8R7: Mathematical Modeling			
Credits &Hours of Teaching		Theory	Practical	Cumulative	
	Credits	4	0	4	
	Hours of Teaching (Total)	60	0	60	
Course Objectives (not more than 100 words)	This course is designed to introduce students to the fundamental principles and techniques of mathematical modeling, focusing on applications in biology, ecology, and epidemiology. Students will explore the role of mathematics in problem-solving and the modeling process, including the development of elementary mathematical models. The course covers modeling with ordinary differential equations (ODEs) and difference equations, examining concepts such as stability, eigenvalues, and fixed points. By the end of the course, students will be able to apply these mathematical tools to analyze real-world problems in population dynamics and related fields.				
Units	Course Content				Teaching Hrs
I	Introduction to Mathematical Modeling: modeling process, Elementary mathematical models; Role of mathematics in problem solving.				12
II	Single species population model: The exponential model, logistic model, harvesting model, and its critical value.				12
III	Modeling with Ordinary Differential Equations: Overview of basic concepts in ODE and stability of solutions: steady state and their local and global stability; Some applications in Epidemiology and ecology.				12
IV	Modeling with Difference Equations: Overview of basic concepts concerning matrices.				12
V	Eigenvalues and Eigenvectors; Fixed points, stability, and iterative processes; Applications to population growth				12
Texts / References	1. J.N. Kapur: Mathematical Modeling, Wiley Eastern, 1988. 2. D.N. Burghes: Mathematical Modeling in the Social Management and Life Science, Ellie Herwood and John Wiley, 1980. 3. F. Charlton: Ordinary Differential and Difference Equations, Van Nostrand, 1965.				
Learning Outcomes (not more than 100 words)	By the end of this course, students will Understand the process of mathematical modeling and its application in solving real-world problems. Develop and analyze models using ordinary differential equations, focusing on stability and steady-state solutions. Apply difference equations, including matrices, eigenvalues, and eigenvectors, to model population dynamics. Gain insight into the stability and iterative processes of mathematical models. Utilize mathematical models to explore and predict outcomes in epidemiology and ecology.				

Course Title		MATMJ88/MATMJ8R8: Numerical Solutions of Partial Differential Equations			
Credits & Hours of Teaching		Theory	Practical	Cumulative	
	Credits	3	1	4	
	Hours of Teaching (Total)	45	30	75	
Course Objectives (not more than 100 words)	This course aims to equip students with the knowledge and skills required to numerically solve partial differential equations (PDEs). It covers the numerical solutions of parabolic, hyperbolic, and elliptic PDEs in various coordinate systems and dimensions, emphasizing explicit and implicit schemes. Students will learn about convergence, stability analysis, and advanced methods like alternating direction implicit (ADI) techniques. Additionally, the course introduces the finite element method, focusing on linear, triangular, and rectangular elements. By the end of the course, students will be able to apply these numerical methods to solve complex boundary value problems.				
Units	Course Content				Teaching Hrs
I	Numerical solutions of parabolic PDE in one space: two and three levels explicit and implicit difference schemes. Convergence and stability analysis.				9
II	Numerical solution of parabolic PDE of second order in two space dimension: implicit methods, alternating direction implicit (ADI) methods. Nonlinear initial BVP.				9
III	Difference schemes for parabolic PDE in spherical and cylindrical coordinate systems in one dimension.				9
IV	Numerical solution of hyperbolic PDE in one and two space dimension: explicit and implicit schemes. ADI methods. Difference schemes for first order equations.				9
V	Numerical solutions of elliptic equations, approximations of Laplace and biharmonic operators. Solutions of Dirichlet, Neuman and mixed type problems. Finite element method: Linear, triangular elements, and rectangular elements.				9
Texts / References	1. M.K. Jain; S.R.K. Iyenger; R.K. Jain: Computational Methods for Partial Differential Equations, Wiley Eastern, 1994. 2. M.K. Jain: Numerical Solution of Differential Equations, 2nd edition, Wiley Eastern. 3. S.S. Sastry: Introductory Methods of Numerical Analysis, , Prentice-Hall of India, 2002. 4. D.V. Griffiths; I.M. Smith: Numerical Methods of Engineers, Oxford University Press, 1993. 5. C.F. General; P.O. Wheatley: Applied Numerical Analysis, Addison- Wesley, 1998.				
Learning Outcomes (not more than 100 words)	Upon completing this course, students will: Understand and implement numerical methods for solving parabolic, hyperbolic, and elliptic PDEs in different coordinate systems. Analyze the convergence and stability of explicit and implicit difference schemes. Apply advanced techniques like ADI methods to multi-dimensional PDE problems. Solve boundary value problems using appropriate numerical methods, including Dirichlet, Neumann, and mixed-type problems. Utilize the finite element method for solving PDEs, focusing on linear, triangular, and rectangular elements, and understand their application to real-world problems.				

Course Title	MATMJ89/MATMJ8R9: Riemannian Manifolds				
Credits & Hours of Teaching		Theory	Practical	Cumulative	
	Credits	4	0	4	
	Hours of Teaching (Total)	60	0	60	
Course Objectives (not more than 100 words)	This course introduces students to the fundamental concepts of Riemannian geometry, focusing on Riemannian metrics, connections, and the geometry of curved spaces. The course covers essential topics such as geodesics, curvature tensors, and Jacobi fields. Students will explore theorems like Hopf-Rinow, Hadamard's, and Cartan's and study complete manifolds and real space forms. Additionally, the course examines the fundamental equations governing Riemannian submanifolds.				
Units	Course Content				Teaching Hrs
I	Riemannian Metrics. Riemannian connection. Fundamental Theorem of Riemannian Geometry. Geodesics.				20
II	Curvature - Riemann curvature, Sectional curvature, Ricci curvature, scalar curvature. Kulkarni-Nomizu tensor fields. Conformal curvature tensor field.				20
III	Jacobi Fields. Complete manifolds, Hopf-Rinow Theorem, The Theorem of Hadamard. Real space forms, Theorem of Cartan on the determination of the metric by means of the curvature. Fundamental equations for Riemannian submanifolds.				20
Texts / References	1. S.S. Chern; W.H. Chen; K.S. Lam: Lectures on Differential Geometry, World Scientific, 2000. 2. M.P. do Carmo: Riemannian Geometry, Birkhauser, 1992. 3. N.J. Hicks: Notes on Differential Geometry, Von Nostrand, 1965. 4. P. Petersen: Riemannian Geometry, Springer 2006. 5. J. Jost: Riemannian Geometry and Geometric Analysis (6ed. Springer, 2011).				
Learning Outcomes (not more than 100 words)	By the end of this course, students will; understand and apply the concepts of Riemannian metrics, connections, and the fundamental theorem of Riemannian geometry; analyze geodesics and study various curvature forms, including Riemann, sectional, Ricci, and scalar curvatures; utilize theorems like Hopf-Rinow and Hadamard to explore the properties of complete manifolds; grasp the significance of Jacobi fields in understanding the behavior of geodesics; apply the fundamental equations of Riemannian submanifolds to solve geometric problems.				

Course Title		MATMJ810/MATMJ8R10: Information and Coding Theory			
Credits &Hours of Teaching		Theory	Practical	Cumulative	
	Credits	4	0	4	
	Hours of Teaching (Total)	60	0	60	
Course Objectives (not more than 100 words)	This course aims to provide a comprehensive understanding of Shannon's information theory, including source and channel coding fundamentals. Students will explore concepts like uniquely decodable and instantaneous codes, entropy, and mutual information alongside key inequalities and coding techniques such as Huffman and Shannon-Fano coding. The course also delves into error-correcting codes, covering topics like Hamming distance, linear codes, decoding rules, and bounds. Advanced topics include perfect codes, the Hamming and Golay codes, and MDS codes.				
Units	Course Content				Teaching Hrs
I	Overview of Shannon theory, Source coding and Channel coding, Uniquely decodable codes, Instantaneous codes, Kraft's and McMillan's inequality, Binary Huffman codes, Average wordlength of Huffman codes, extensions of sources				12
II	Information and entropy, properties of the entropy function, entropy and average word length, Shannon-Fano coding, Shannon's first theorem, Binary symmetric channel, system entropies, Mutual Information and channel capacity				12
III	Error correcting codes: maximum likelihood decoding, Hamming distance and distance of a code, minimum distance decoding rule, Linear codes, minimum distance of linear codes, Hamming weight, Bases for linear codes, Generator matrix and parity-check matrix, equivalence of linear codes, (Slepian) standard array and syndrome decoding				12
IV	Hamming's sphere-packing bound, the Gilbert-Varshamov bound. Perfect codes, The Hamming codes and the Golay codes, Singleton bound and MDS codes, Hadamard matrices and codes.				12
Texts / References		1. G.A. Jones; J.M. Jones: information and Coding Theory: Springer, 2000. 2. S. Ling; C.Xing: Coding Theory, Cambridge University Press, 2004. 3. J.H.V. Lint: Introduction to Coding Theory, Springer, 1982.			
Learning Outcomes (not more than 100 words)		By the end of the course, students will understand the principles of Shannon theory, source and channel coding, and the mathematical foundation of entropy; analyze and construct optimal coding schemes, including Huffman and Shannon-Fano codes; apply concepts like mutual information and channel capacity to evaluate communication systems;gain proficiency in error-correcting codes, including linear codes, decoding methods, and bounds. develop the ability to design and evaluate advanced codes, such as Hamming, Golay, and MDS codes, for reliable data transmission.			

Course Title	MATMJ811/MATMJ8R11: Algebraic Geometry				
Credits & Hours of Teaching		Theory	Practical	Cumulative	
	Credits	3	0	3	
	Hours of Teaching (Total)	45	0	45	
Course Objectives (not more than 100 words)	This course aims to introduce the students to the topic of algebraic geometry. This course, as the name suggests is a fusion of Algebra and geometry. The students would have already had familiarity with algebra and geometry and in this course they will see how the two subjects of Mathematics come together to help each other.				
Units	Course Content				Teaching Hrs
I	Introduction and Overview, Affine Space, Affine plane curve, Projective space, projective space curve, their properties using set theoretic operations, Examples.				9
II	Affine algebraic sets, Algebraic sets in terms of zeros of polynomials, Projective Algebraic sets, Zariski Topology, irreducible space, Noetherian space, affine Variety and quasi affine variety, Dimension, Krull dimension				9
III	Interlude to field theory, extension, irreducible and minimal polynomial, Algebraically closed field,, Ring extension, integral element, module finiteness, integrally closed extension.				9
IV	Weak nullstellensatz, Hilbert's Nullstellensatz and their proof, Transcendence base, Polynomial map				9
V	Rational functions, Coordinate ring, valuations, completion, topological field, Tangent space, Derivations				9
Texts / References	1. D Mumford: The red book of varieties and schemes, Springer, 1999. 2. J S Milne: Algebraic Geometry, Web version, 2023. 3. S S Abhyankar: Algebraic Geometry for Scientists and Engineers, AMS, 2011. 4. Igor R. Shafarevich: Basic Algebraic Geometry 1, Springer, 1994.				
Learning Outcomes (not more than 100 words)	After completion of this course the students will have a detailed account of foundation of Algebraic Geometry. The student will also be able to appreciate that the problems of geometry have also been approached via other methods which are different from polynomial approaches. After attending this course a student will be in a better position to approach the advanced topics in Algebraic geometry.				